

Knowledge is Power: A Field Experiment in the Chinese and US Stock Markets

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Abstract

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JEL Classification: G14; G41, G53, C93

Keywords: Financial knowledge, accrual mispricing, investor education, field experiment, social media

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Abstract

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1. Introduction

“Knowledge is power. Information is liberating. Education is the premise of progress, in every society, in every family.”

— Kofi Annan

Having knowledge of financial concepts and skills is crucial for making sound financial decisions. Research indicates that financial knowledge influences individuals’ financial behavior in various aspects, such as market participation and asset allocation.¹ Given that financial behavior drives asset prices, it is surprising that the literature does not directly assess the effects of financial knowledge on asset prices. This study aims to fill this gap by examining whether financial knowledge affects asset pricing. Education can empower investors with the knowledge and skills needed to process and interpret public information (e.g., corporate disclosures). This, in turn, supplies the market with more informed investors, improving the efficiency of stock prices.

This topic poses both conceptual and empirical challenges. First, in perfect market conditions, where investors possess comprehensive knowledge and the full capacity to process information, the addition of education may not influence their decisions and, consequently, stock pricing (Simon, 1987). In reality, investors have limited cognitive resources and imperfect knowledge (Kahneman and Tversky, 1973; Simon, 1955). Education may still prove ineffective if investors lack motivation and the requisite expertise to comprehend educational materials (Fernandes et al., 2014; Piaget, 2003). Furthermore, there is an overarching challenge of ensuring that investors apply the knowledge they have gained in real decisions (Merkoulova and Veld, 2022).²

Empirically, it is hard to establish a causal link between financial knowledge and asset pricing. One primary hurdle is the potential endogeneity between financial knowledge and stock prices (Bianchi, 2018; Lusardi and Mitchell, 2014). For instance, financial knowledge and market outcomes can be simultaneously shaped by less observable factors such as individuals’ time preferences and wealth endowment (Jappelli and Padula, 2013; Meier and Sprenger, 2013).

¹ The literature shows that individuals with low financial literacy are more likely than others to make poor financial decisions (Bianchi, 2018; Von Gaudecker, 2015), avoid diversification (Merkoulova and Veld, 2022), hold risky assets with low expected returns (Bianchi, 2018), discount professional financial advice (Cole et al., 2011), hold lower risk-adjusted stock returns (Von Gaudecker, 2015), avoid planning for retirement (Song, 2020), hold insufficient savings (Cole et al., 2011), default on mortgage payments (Gerardi et al., 2013), and have lower credit scores (Brown et al., 2019). Please see Lusardi and Mitchell (2014) for a review.

² For example, individuals may be plagued by certain personal traits that are inherited and hard-wired (Altman, 2012) or are dictated by a certain institutional and incentive environment (Posner, 2009).

Moreover, there is a possibility of a feedback loop: when investors gain experience and learn from past stock prices, their financial knowledge deepens, which in turn could influence prices.

To tackle these challenges and investigate our research question, we conduct a field experiment that involves educating investors about accounting accruals, a concrete type of financial knowledge. Accruals represent the difference between a firm's reported earnings and its underlying cash flows, reflecting the firm's estimates of future benefits and obligations. In essence, the accrual component of earnings is less persistent than the cash flow component in predicting future earnings. However, numerous studies show that, due to cognitive limitations, investors often overlook the transient nature of accruals and overprice stocks with high accruals, a phenomenon known as the "accrual puzzle" (Allen et al., 2013; Chan et al., 2006; Richardson et al., 2005; Sloan, 1996; Xie, 2001). Our study investigates whether investor education can address the puzzle.

We choose to focus on accruals for three reasons. First, the literature identifies accrual mispricing as one of the most pervasive and robust financial anomalies (Avramov et al., 2013; Fama and French, 2008; Hirshleifer et al., 2012). While extensive efforts are made to explain this anomaly (Khan, 2008; Mashruwala et al., 2006), the mechanisms underlying accrual mispricing remain unclear (Richardson et al., 2010). Second, unlike basic financial concepts such as compounding, inflation, interest rates, and diversification (Lusardi and Mitchell, 2014; Van Rooij et al., 2011), accruals represent a more intricate concept. Gaining proficiency in this area can notably enhance an individual's financial acumen (Lusardi and Mitchell, 2014). Third, given that grasping the economic implications of accruals demands a sophisticated cognitive aptitude (Collins et al., 2003), accruals present an ideal base to test the efficacy of investor education.

We implemented randomized controlled trials (RCTs) to examine whether providing accrual knowledge to investors reduces accrual mispricing. Our experiment has two distinctive features. First, we conveyed this specialized knowledge to investors via social media platforms. Evidence indicates that social media, serving as vital conduits of information for investors, play a significant role in influencing investor behaviors and stock market outcomes (Miller and Skinner, 2015). A testament to this is the stock price rebound for GameStop and AMC in January 2021, highlighting the growing sway of social media in the investment community.³

³ See <https://www.investmentnews.com/gamestops-saga-what-the-stock-surge-proves-about-financial-literacy-205019>.

Second, we conducted our experiment to encompass firms listed in both the Chinese and US markets. Nobel laureate Abhijit Banerjee posits that “a very simple approach to generalization ... is to implement the same concept in multiple locations to build confidence in its impact” (Banerjee, 2020, , p.1938). As the leading developed market, the US operates under rigorous securities regulations and features significant participation of institutional investors. Conversely, China, while being the world’s largest emerging market and second-largest stock market by capitalization, has a more relaxed regulation environment. Additionally, the Chinese market is dominated by retail investors, who drive nearly 90% of daily trades (Titman et al., 2022). Given the large institutional differences between the two countries, any consistent findings from our study would support the generalizability of our conclusion.

Our experiment ran from January to December 2020, covering all publicly listed firms in China and the US. We developed two distinct educational modules on the concepts and pricing implications of accruals. The first module offers a conceptual introduction of accruals, detailing their definition, properties, and implications for future earnings (conceptual knowledge). The second module introduces the methodologies for estimating accruals, detailing specific models, key variables, data sources, and tools (methodological knowledge). The materials are made available in Chinese and English for the respective markets.

We disseminate these materials using social media platforms in each country, namely Weibo, Xueqiu, and Guba EastMoney in China and Twitter and StockTwits in the US. Over a 19-day window around firms’ earnings announcement (EA) days in 2020, we share the two sets of knowledge, along with the upcoming EA news, on the designated tag pages for the *treatment group* of stocks. For a subset of stocks, only the conceptual knowledge and EA news are shared. Notably, each post contains a hyperlink directing users to a custom webpage for each stock on a website that we established, ensuring easy access to the complete educational materials.

We create a *control group* of stocks by merely sharing the upcoming EA news as a placebo. In theory, any observed difference in accrual pricing between the *treatment* and *control groups* should reflect the effect of accrual education. Additionally, we created a *comparison group* of stocks for which we did not share any information. The difference between the *treatment* and *comparison groups* quantifies the overall impact of our intervention. Our final dataset comprises 2,284 stocks from the Chinese market and 2,387 stocks from US market. Our educational materials

garnered a total of 26,834,089 views in China and 3,397,437 views in the US.⁴ Diagnostic tests indicate no significant difference in firm characteristics between the treatment and control groups. The stock price movement prior to our experiment is also parallel across the groups.

Following the approach of Sloan (1996), we delve into the pricing of accruals announced during our experiment period across the sample groups in each country. Our findings confirm that accruals exhibit lower earnings persistence than cash flows across all of the groups. In the control and comparison groups, stock prices do not seem to rationally account for the low earnings persistence of accruals relative to cash flows, pointing to a mispricing of accruals. However, this mispricing diminishes in the treatment groups in both markets, especially among those exposed to both conceptual and methodological knowledge. This underscores the effectiveness of our education treatments in reducing accrual mispricing.

The mispricing of accruals can be broken down into two parts: accruals being overvalued during EAs and a subsequent price correction in the post-EA phase. We next explore these two elements. First, we examine whether our treatments reduce accrual overpricing at EAs. Through a difference-in-differences (DID) analysis, we find that EA returns for stocks with high accruals are lower in the treatment group than in the control group in both markets, suggesting that the education treatment leads to a price discount for stocks with high accruals. Specifically, relative to control stocks, a one-standard-deviation increase in accruals for treatment stocks is, on average, associated with a 0.3% (0.8%) reduction in the two-day abnormal return in China (the US). Notably, the combined effect of providing both conceptual and methodological knowledge is more pronounced than the effect of offering only conceptual knowledge. This indicates that introducing methodological knowledge adds value beyond what is achieved with conceptual understanding.

Second, we examine whether our treatments mitigate subsequent price reversal. As expected, we find that portfolios that buy low accrual stocks and sell high accrual stocks exhibit significant negative abnormal returns in the year post-EAs in the control groups. This pattern is absent in the treatment groups. The DID analysis reveals that a one standard deviation increase in accruals is associated with a 3.2% (6.1%) decline in buy-and-hold abnormal returns in the following year in China (the US). The reduction becomes negligible for treatment stocks. Again, the treatment effect is most evident in the group exposed to both conceptual and methodological knowledge.

⁴ In the US, read counts come solely from Twitter, as StockTwits does not provide such data.

We obtain similar results when using the comparison group as the benchmark group. Additionally, a placebo test based on historical accruals yields no comparable treatment effects, suggesting that our results are falsifiable. Furthermore, we find that the treatment effects are more pronounced when the investors engage more with the educational materials (e.g., increased reading, comments, retweets, likes, and saves on our posts, as well as more clicks on the hyperlinks within the posts) and for stocks predominantly owned by retail investors.

We conduct tests to address alternative explanations. First, no education effect is found when comparing the control and comparison groups, negating the impact of treatment salience. Second, the education effect is evident only for positive accruals, aligning with our education's focus on high accrual implications and refuting the notion that our exercises merely amplify attention to accruals. Third, the education effect persists one year post-experiment without new message postings, suggesting that our treatment effect is not fear-induced. Fourth, the effect remains pronounced in stocks with high trading turnover, implying a limited impact from the spillover effect due to common ownership between the treatment and control stocks. Lastly, our results stay robust even after accounting for the fixed effect of EA months and excluding EAs during the pandemic peak, indicating minimal distortion from the COVID-19 outbreak.

First, our study contributes to the literature on the effects of financial knowledge in financial markets. Existing studies mainly focus on the effects of financial knowledge on individuals' financial decisions and behaviors (Lusardi and Mitchell, 2014). To the best of our knowledge, ours is the first work to directly test the influence of financial knowledge on asset pricing.

Second, our study extends research that examines financial anomalies and pricing efficiency. Notably, McLean and Pontiff (2016) posit that the academic discovery of financial anomalies can enhance market efficiency by enabling investors to pinpoint arbitrage opportunities. Our research underscores that not only is the discovery of such knowledge vital, but its dissemination and the education of investors regarding it are also crucial for improving market efficiency.

Third, previous studies on accrual anomaly rest on explanations such as risk (Chichernea et al., 2015; Khan, 2008), transaction costs (Collins et al., 2003; Mashruwala et al., 2006), limited attention (Hirshleifer et al., 2011; Miao et al., 2016), and insufficient supplementary accounting information (Louis et al., 2008; Miao et al., 2016). We extend this body of literature by demonstrating a new explanation, namely investors' limited understanding of accounting accruals. This perspective also sheds light on the understanding of other price anomalies.

Lastly, our research carries policy implications. Individuals in many developed and developing countries have low levels of financial literacy (Merkoulova and Veld, 2022; Van Rooij et al., 2011). Although many financial education programs have been introduced worldwide to rectify this problem (Kaiser et al., 2022), their efficacy remains debatable (Lusardi and Mitchell, 2014). Our findings underscore the potential of social media platforms as tools for investor education. Furthermore, our results highlight the importance of integrating methodological applications with conceptual explanations in financial education. This insight holds considerable relevance for the design and execution of financial education programs.

2. Literature and hypothesis development

2.1. Related literature

“Financial knowledge,” or “literacy,” refers to people’s ability to process economic information and make informed financial decisions (Lusardi and Mitchell, 2014). Individuals’ accumulation of financial knowledge can affect the development of financial markets. Cole et al. (2011) note that limited financial literacy presents a significant barrier to demand for financial services. Supporting this view, Van Rooij et al. (2011) find that individuals with low financial literacy are less likely than others to buy stocks and more likely to rely on informal channels for financial advice.

Moreover, financial knowledge affects individuals’ investment performance. For example, Von Gaudecker (2015) finds that investors with high levels of financial literacy receive higher risk-adjusted returns. In their study exploring the mechanism underlying the relationship between financial literacy and returns, Bianchi (2018) notes that more literate investors actively rebalance their portfolios and tend to hold riskier positions that have higher expected returns and are less likely to engage in trend-chasing behaviors than others.

Financial knowledge also affects individuals’ debt-related choices. For example, Klapper et al. (2012) find that individuals with higher financial literacy are less likely to use informal sources of borrowing and have a greater ability to cope with macroeconomic shocks than their less financially literate peers. Moreover, individuals with low financial literacy are less likely than others to plan for retirement (Song, 2020) and accumulate savings (Cole et al., 2011) and more likely to default on mortgage payments (Gerardi et al., 2013). Given the substantial costs of financial illiteracy, Lusardi et al. (2017)’s finding that 30%–40% of wealth inequality is attributable to financial knowledge is not surprising.

Collectively, the literature focuses on the effects of basic financial concepts on individuals' financial behavior and decisions. Yet, few studies directly link more concrete financial knowledge to the pricing of assets in financial markets. We seek to fill this gap in the literature.

2.2. Hypothesis development

Accruals are fundamental to earnings measurements and represent firms' estimates of future benefits and obligations.⁵ Accountants use accruals to recognize the financial effects of transactions when they become probable, rather than when the cash consequences are realized. For example, a positive (negative) accrual occurs when there is an increase in account receivables (payables). Accruals reverse when either (i) the expected future benefits are realized (e.g., account receivables are collected and payables are paid off) or (ii) there is evidence indicating that future benefits are unlikely to materialize (e.g., inventories are written down if they are obsolete).⁶ As a result of measurement errors or the possible occurrence of case (ii), accruals have lower earnings persistence than cash flows. In other words, the current level of accruals has a weaker power than the current level of cash flows in predicting future earnings.

Given the inherent uncertainty and complexity associated with accruals, they pose a significant challenge for investors aiming to make informed decisions. To traverse this intricate landscape, accurately pricing accruals demands two costly cognitive undertakings: investors should both conceptually understand the distinctive implications of the accrual and cash flow components of a given company's reported earnings and methodologically apply formulas and models to estimate the level of accruals and construct measures to gauge their quality.

A large body of literature suggests that investors fail to process accrual information correctly and efficiently. Many investors seem not to understand the low earnings persistence of accruals relative to cash flows and, consequently, fixate on reported earnings (Allen et al., 2013; Chan et al., 2006; Richardson et al., 2005; Sloan, 1996; Xie, 2001). As such, high accruals are overpriced and negatively related to future stock returns. This accrual mispricing is often attributed to investors' cognitive limitations.⁷ For example, Sloan (1996), who first recognized this mispricing,

⁵ See our education materials in Internet Appendixes 1.1 and 1.2 for a detailed introduction to accruals.

⁶ Case (i) means that accruals correctly anticipate future benefits or obligations. Case (ii) refers to accrual estimation errors (i.e., accruals reverse without cash flow realization), which can be caused either by unintentional estimation errors or reporting manipulation and distortion.

⁷ Studies also explore accrual mispricing based on risk (Khan, 2008) and arbitrage costs (Mashruwala et al., 2006). However, these explanations only provide a partial explanation for accrual mispricing (Richardson et al., 2010).

suggests that it is caused by “the inability of investors to distinguish fully between the different properties of the accrual and cash flow components of earnings” (p. 290).

We propose that educating investors on the implications and measurement of accruals can enhance their cognitive capacity to comprehend and process accrual-related information. Investors often neglect accounting information, either due to cognitive bias or a rational allocation of their limited capabilities (Cohen et al., 2020; Lu, 2022). Both of these tendencies are rooted in the inherent limitation of human cognitive resources (Kahneman, 1973). An introduction to the concept and implications of accruals can stimulate active cognitive learning (Piaget, 2003; VanLehn, 1996), prompting investors to refine their grasp of accruals’ nature and discern the distinct properties of earnings’ cash flow and accrual components. This deepened understanding can alleviate investors’ cognitive load, allowing them to navigate bias and allocate their attention more effectively across various types of information (Hirshleifer and Teoh, 2003). Consequently, investors could be more attentive to accrual information, reducing the likelihood of fixating on bottom-line reported earnings.

Imparting methodological knowledge to investors arms them with the skills to quantify the magnitude and quality of accruals. In certain markets, such as the US, the disclosure of a statement of cash flows (SCF) in firms’ EAs is uncommon (Miao et al., 2016).⁸ Consequently, investors cannot directly derive accruals by subtracting cash flows from net incomes. Instead, they must deduce accruals from the income statement and comparative balance sheets.⁹ Moreover, the earnings persistence of accruals depends on the reliability of the accruals (Chan et al., 2006; Dechow and Dichev, 2002; Richardson et al., 2005). Supporting this, Xie (2001) shows that the primary source of accrual mispricing is abnormal accruals. Therefore, even if investors can derive accruals from SCF, they ought to further develop measures to effectively capture and trade on this mispricing signal. Our methodological educational contents provide the formula and essential variables necessary to compute both accruals and abnormal accruals. Furthermore, engaging in methodological practices may foster a deeper understanding and integration of conceptual knowledge through active participation and reflection (Billett, 2010), which amplifies investors’ learning outcomes.

⁸ D’Souza et al. (2010) report that during their sample period of 2000–2003 in the US, only 14.5% contain SCF items.

⁹ About 90% of US firms had disclosed balance sheet items as of the second quarter of 2012, and this percentage has increased in recent years (Miao et al., 2016).

Moreover, research indicates that when the costs of processing information decrease and the advantages of using the gathered information rise, investors are more inclined to seek out information (Blankespoor, 2019; Kim et al., 2019). Similarly, if accrual education lightens the cognitive load associated with processing accrual data, investors may feel motivated to search for additional information, which further improves their accrual pricing capabilities (Bertrand and Morse, 2011). For example, investors may collect supplementary data on a firm's customers and suppliers to gain a deeper understanding of accrual reliability.

Therefore, with the introduction of such education, investors are likely to become more sophisticated in processing accrual information. When investor sophistication increases in the market, the probability that marginal investors can better price accruals increases, and accrual mispricing is attenuated (Collins et al., 2003; Hand, 1990). Our hypothesis is as follows:

***H1:** Accrual mispricing of stocks whose investors receiving the treatment of investor education is lower than that of control stocks whose investors do not receive the treatment.*

3. Experimental design and data

For our study, we procured information on the concept and pricing implications of accruals from top-tier accounting and finance journals. We then disseminated this knowledge to investors in randomized groups of stocks via social media platforms in China and the US. We explain the design and execution of the experiment in detail in the following subsections.¹⁰

3.1. Educational materials

To disseminate knowledge to investors, we established a website, <http://www.financial-education-hub.com>. On this site, we dedicated a unique webpage to each stock, each containing details on either the conceptual or methodological knowledge of accruals. The pages are in Chinese (English) for Chinese (US) firms.

All of the stock webpages share the same conceptual knowledge content on accruals, samples of which are available in Internet Appendix 1.1. Each article begins with a “highlight” and a brief “example,” aiming to provide investors with a quick understanding of the basic concept of accruals and the associated pricing implications for the firm's future performance. The body of the article includes seven parts, including the definition of “accruals,” the reasons for the use of accruals in

¹⁰ We obtained human ethics approval from our university to conduct the experimental exercise in this project.

accounting, the relationship between accruals and cash flow realization, the implications of accruals on firms' future performance, potential factors that affect the low earnings persistence of accruals, other reasons for low future profitability following high accruals, and a summary. A list of references is also included at the end of the article. For enhanced comprehension, we incorporated an animated video summarizing the article, positioned right after the highlight.¹¹

The methodological content on accruals is tailored for each stock, samples of which are presented in Internet Appendix 1.2. A link redirects readers to the conceptual article, ensuring that the basic concept of accrual can be grasped before delving deeper. To aid in the application of this knowledge, the webpage features a spreadsheet tool to compute a firm's accruals, both normal and abnormal. The tool provides variables, demos, and accrual implications. Historical firm data is pre-loaded into the spreadsheet, requiring readers only to input the most recent figures to get results.

Samples of the spreadsheet and results are provided in Internet Appendix 1.3. The results display a line chart tracing the firm's earnings, accruals, and abnormal accruals since 2009. It highlights the current year's accrual level and the ten-year average. Readers can gauge a firm's accrual quality by examining the current year's abnormal accrual level and its ten-year volatility.¹² We also offer the firm's ranking for each metric as compared to its industry peers and the market.

3.2. Experimental sample

3.2.1 Experiment period

The experiment period in China took place between January and May 2020, during which Chinese firms' annual financial reports for 2019 were released.¹³ The experiment period in the US was from January to December 2020, during which US firms' annual performance for the fiscal year of 2019 (2020) was announced, if the firms' fiscal year ended during January to May (June to December). For simplicity, hereafter, we refer to the annual earnings that were announced in 2020 as earnings in 2019 for both markets.

3.2.2 Sample firms

¹¹ Prior studies show that the use of visuals improves individuals' learning (Nekrasov et al., 2022).

¹² The abnormal accruals reported in the spreadsheet are estimated at the individual firm level based on the model of Dechow and Dichev (2002).

¹³ The fiscal year end for firms listed in China is December, and all of the firms listed on China's two stock exchanges are required to file their annual performance from January to May of the following year.

The initial experimental subjects include all of the stocks that were publicly listed in China and the US by the end of 2018, the last year for which data on annual performance were available before our experiment. We exclude firms in the financial services industries because they are subject to different regulations and their disclosure practices differ from those of firms in other industries (Fang et al., 2015; Firth et al., 2019). We also exclude stocks having fewer than four observations from 2009 to 2018.¹⁴ We further exclude stocks with no trading information in 2018. The final number of unique stocks in our experiment is 2,308 in China and 2,547 in the US. The filtering process used in constructing the sample is reported in Appendix 1.

3.2.3 *Treatment assignment*

We created four randomized groups of stocks in each country.¹⁵ We provided the upcoming EA news and the conceptual knowledge to investors for the stocks in the first treatment group, T₁. We provided the upcoming EA news and both the conceptual and methodological knowledge to investors for the stocks in the second treatment group, T₂. We provided the upcoming EA news, as a placebo, to investors for the stocks in the control group, C. Therefore, the difference in accrual pricing between T₁ (T₂) and C reflects the effect of the treatment of accrual knowledge. The final group is a comparison group, S. We did not make any intervention regarding the stocks in this group to establish this group as an alternative benchmark. The difference in accrual pricing between this group and the treatment groups quantifies the overall impact of our intervention.

Appendix 1 details the distribution of firms across these groups. In China, each group comprises 577 stocks. In the US, group S contains 636, while each of the other groups has 637 stocks. We conduct a comparative analysis of these groups based on the firms' characteristics from the year preceding the experiment and present the results in Table 1. Validating our randomization process, the analysis reveals no significant differences between the groups across all of the metrics.

[Insert Table 1 about here]

3.3. *Experimental execution*

¹⁴ To estimate firm-level abnormal accruals as in Dechow and Dichev (2002), a minimum of four observations is required.

¹⁵ We ranked stocks by return on assets in FY2018 within each of the two stock exchanges (SHSE and SZSE) in China and the three exchanges (NYSE, Nasdaq, and Amex) in the US. Every fourth stock within each market was selected as a stock in a designated group, leaving the stocks in each of the two markets divided evenly among four groups. We rank the stocks based on return on assets, as return on assets, or earnings, is the primary factor to control.

3.3.1 *Knowledge dissemination platforms*

We used social media platforms to share the knowledge contents with investors. In China, we used Weibo, Xueqiu, and Guba EastMoney, while in the US, we used Twitter and StockTwits.¹⁶ Twitter and Weibo, which are major global and Chinese social media platforms, respectively, allow users to post short messages and use “cashtags” to link to specific companies (e.g., \$AAPL is linked to the page of Apple Inc.). StockTwits, a niche platform for stock investors, is akin to China’s Xueqiu but allows longer posts and company-specific tagging. Guba EastMoney, China’s leading stock message board, primarily disseminates financial news.

3.3.2 *Constructing social media posts*

We crafted social media posts summarizing the key points of accrual knowledge, embedding hyperlinks to direct readers to the full page of the knowledge information.¹⁷ Initial samples of these posts, both in English and Chinese, are available in Internet Appendix 2. The opening sentence of each post is about the EA news. The term “\$Ticker” represents a stock’s trading symbol. We provided two distinct URL links: one leading to the detailed conceptual knowledge of accruals (type 1 page) and the other to the detailed methodological knowledge of accruals (type 2 page), depending on what test group the associated stock belongs to.

Specifically, the stocks in group T₁ are linked to the type 1 page, while those in group T₂ are linked to both type 1 and type 2 pages. As an example, Coca-Cola (ticker: COKE) from group T₂ has two pages: a type 1 page (<http://www.financial-education-hub.com/1/COKE>) and a type 2 page (<http://www.financial-education-hub.com/2/COKE>). In contrast, Google (ticker: GOOG) from group T₁ only has a type 1 page (<http://www.financial-education-hub.com/1/GOOG>). No pages are assigned to the stocks in groups C and S. It is worth noting that platforms like StockTwits, Guba EastMoney, and Xueqiu restrict hyperlink usage. For these platforms, we opted to publish

¹⁶ Social media platforms have become crucial sources of information for investors (Jame et al., 2016; Lee et al., 2015). Recent studies show that investors are able to obtain useful information from social media platforms such as Twitter (Bartov et al., 2018; Blankespoor et al., 2014; Tang, 2018), StockTwits (Cookson and Niessner, 2020), and Weibo (China’s Twitter) (Feng and Johansson, 2019).

¹⁷ Social media platforms usually limit the number of characters in each post. Hyperlinks are widely used to direct investors to pages featuring more thorough content. Blankespoor et al. (2014) find that firms’ bid–ask spread is reduced when firms use Twitter to send links to press releases to market participants.

full-length articles directly on the stock tag pages to avoid the use of hyperlinks.¹⁸ To enhance engagement and avoid redundancy, we ensured that each post was uniquely phrased.

3.3.3 *Sending social media posts*

We sent social media posts to investors in the designated groups of stocks during a 19-day window around the firms' EA dates. The initial communication schedule is provided in Internet Appendix 3. We posted seven, one, and seven social media posts before, on, and after the EA day, respectively. In total, we aimed to share 15 posts for each firm on each social media platform. However, as some of the platforms do not allow users to share similar information within a certain period, we adjusted our posting frequency on such platforms.¹⁹

We created three types of accounts with fictional names on each platform. Each type of account only posts information that is targeted to one of the groups of stocks. Moreover, to avoid unnecessary duplicate posts, we created new posts only when the previous post had been buried.²⁰

3.3.4 *Collecting experimental data*

We tracked various metrics, such as the number of reads, comments, retweets, likes, reshares, and saves, one day after each post to gauge the reach and engagement of our content. We completed the experiment by the end of May (December) 2020, when our sample had 2,284 (2,387) stocks with valid information in the Chinese (US) markets (see Appendix 1).

Table 2 showcases our outreach efforts. In China, we dispatched 54,734 posts, which garnered approximately 26.8 million reads, 13,060 comments or replies, 13,704 retweets or reshares, 808 likes (from Xueqiu and Weibo), and 392 saves (from Xueqiu). Further daily breakdowns and sample responses are available in Internet Appendices 4 and 5, respectively. Our website attracted 10,896 visits from Weibo users. For platforms like Guba EastMoney and Xueqiu, we presented full articles instead of hyperlinks; therefore, their traffic is not reflected in these figures.

¹⁸ On platforms with word limits for posts, we used images instead of text. Specifically, we posted tweets accompanied by screenshots of our web pages.

¹⁹ The specific adjustment varies from platform to platform, depending on the severity of the restrictions. In practice, we reduced our communication frequency from the above-described daily frequency to every other day for Guba EastMoney (11 posts per firm in total), every three days for Xueqiu (five posts per firm in total), and every five days for StockTwits (three posts per firm in total). We did not make any adjustments to our Twitter and Weibo posts, as these platforms are the least restrictive about users' posting activities.

²⁰ I.e., when the most recent 20 posts on the stock's tag page do not belong to us or when our most recent post no longer appears on the first page of the stock's tag page due to newer posts.

In the US, we posted 28,429 times, resulting in approximately 3.4 million reads on Twitter, 2,680 comments or replies, 2,321 retweets or reshares, and 3,937 likes. Notably, our outreach was more extensive in China, aligning with the greater presence of retail investors there than in the US (Titman et al., 2017). Our website registered 15,030 visits, exclusively from Twitter, as we use full articles on StockTwits instead of hyperlinks.

[Insert Table 2 about here]

3.4. Data and key variables

To evaluate the treatment effects, we collect financial data for the Chinese firms from the China Stock Market & Accounting Research Database (CSMAR). For the US firms, we adopt the approach of Du et al. (2023) to collect “as-filed” financial statements from the Financial Statement and Notes Data Sets provided by the Securities and Exchange Commission (SEC).²¹ In situations where “as-filed” data for certain items are missing, we supplement them with information from Compustat. We obtain return data from the Center for Research in Security Prices (CRSP).

Our measure of accruals is *Accruals*, which is a firm’s total or normal accruals. Following Sloan (1996), Richardson et al. (2005), and Chan et al. (2006), we define the measure as the change in current assets minus the change in cash holding minus the change in current liabilities plus the change in short-term debt plus depreciation, scaled by total assets. Studies suggest that the mispricing of total accruals mainly stems from unexpected discretionary accruals (Xie, 2001). As such, we also construct a variable for unexpected discretionary accruals ($Accruals_{UD}$), which is discretionary accruals in the current year minus discretionary accruals in the previous year (Balsam et al., 2002; Louis et al., 2008). Specifically, following Kothari et al. (2005), we regress *Accruals* on the reciprocal of total assets; the change in sales/total assets; net property, plant, and equipment/total assets; and the return on assets in each industry-year, and we use the residuals as the measure of discretionary accruals.

Our measure of earnings (*Earnings*) is operating income before depreciation, scaled by total assets. We use the difference between *Earnings* and *Accruals* to measure cash flows (*Cash Flow*)

²¹ Du et al. (2023) suggest that Compustat often adjusts reported financial data for the sake of consistency, which can result in misclassifications or omissions. In contrast, “as-filed” data are more detailed and not subject to proprietary adjustments by data aggregators. Du et al. (2023) evaluate the attributes of accrual accounting using both datasets. Their findings contrast sharply with earlier studies that rely on Compustat data; specifically, they discover that accruals still possess predictive power for future cash flows and that the accrual anomaly remains significant when using “as-filed” data.

(Sloan, 1996). Moreover, we construct the measure of EA surprise (SUE), which is defined as the announced earnings per share minus the consensus analyst forecasts in the past year, scaled by the stock price before the EA.²²

We calculate abnormal returns around the announcement of 2019 earnings to evaluate the market response to our treatment. We use the two-day cumulative abnormal return from day 0 (EA day) to day 1 based on the market model to measure the market reaction to the EA ($CAR(0,1)$). We use the buy-and-hold return from day 11 to day 251 minus the returns of the corresponding 5×5 market value and book-to-market (Size/BM) matched portfolio to measure the future long-term abnormal return after the EA ($CAR(11,251)$). We start from day 11 because we stop posting on day 10, and we use $CAR(11,251)$ to study the future abnormal return after the treatment.

4. Main results

4.1. Preliminary analysis

Recent studies suggest a diminishing presence of the accrual anomaly. For example, Green et al. (2011) find that the hedge abnormal returns, based on Sloan’s (1996) accruals, are not consistently positive after 2004. Hou et al. (2020) extend the sample period to 2016, noting a decline in accrual mispricing. However, its significance persists. Du et al. (2023) show that the attenuation in accrual anomaly is partly due to data discrepancies. They find that the anomaly remains persistent when using as-filed financial statement data instead of Compustat data.

Given the mixed evidence, before moving to our main analysis, we replicate prior studies by running a standard Fama–MacBeth regression based on our full sample in both Chinese and US markets. Specifically, we regress $CAR(11,251)$ on *Accruals* based on an annual panel for the six-year period before the experiment (i.e., from 2013 to 2018). The results, detailed in Internet Appendix 6, show negative and statistically significant coefficients on *Accruals*, verifying the anomaly’s presence in both markets.

Transitioning to our experiment period, we analyze stock reactions to accruals among the four groups of our experiment. Figures 1 and 2 showcase the cumulative abnormal returns (returns minus the returns of the corresponding 5×5 Size/BM matched portfolio) for the stocks with top-

²² If analyst forecasts are not available, we follow Louis et al. (2008) and use the earnings per share of the previous year to proxy for the expected earnings per share.

decile unexpected discretionary accruals ($Accruals_{UD}$) in each market. The group returns are value-weighted based on the market value of equity at the end of 2019.

Figure 1 illustrates the results for the Chinese market. Panel A plots the returns from day -13 to day 13. During the pre-experiment period, we observe similar stock price movements for all of the four groups. The price patterns of groups S and C align during the experiment, both showing an increase. Notably, the prices for group C seem to rise earlier and more sharply than those for group S. This surge is likely to be attributable to our exercises amplifying the focus on EAs, which may spur investor attention. However, a quick reversal occurs on day -1, with prices then trending upward with group S on the EA day. These patterns suggest that the market generally perceives high accruals positively. This points to a mispricing of accruals, irrespective of any intervention.

In contrast, the stock prices for the treatment groups T_1 and T_2 are suppressed. As the experiment commences, their prices trend downward, with returns from day 0 to day 1 being nearly flat. This indicates that the market shows less enthusiasm for news of high accruals for the treatment stocks. It is worth noting that the prices of group T_2 are suppressed more severely than the prices of group T_1 around the EA. This suggests a more pronounced treatment effect when both the conceptual and methodological contents are provided.

Panel B extends the plot to day 251. We find the prices of groups S and C reverse and have a negative return in the post-experiment period. Meanwhile, the prices of group T_1 and T_2 exhibit fluctuations but do not change significantly in terms of cumulative returns. This suggests that accrual mispricing is less manifested in the treatment groups than in the control groups.

Figure 2 plots the results for the US market, where we find a similar pattern. The stock price movement is parallel for the four groups in the pre-experiment period. The prices of groups S and C increase during the experiment period but reverse in the post-experiment period. The prices of group T_1 and T_2 decline during the experiment period but stabilize over the entire period. In contrast to the Chinese market, the prices of group T_2 in the US market decline more substantially during the experiment period. This implies that adding the methodological contents to the conceptual contents has a substantial impact on US investors. Another distinction between the two countries is that in the US, the prices of group S rise more sharply than those of group C. This implies that the attention effect is less pronounced in the US market.

In conclusion, our preliminary data affirm the persistence of accrual mispricing and underscore the efficacy of our educational intervention in addressing this anomaly.

[Insert Figures 1 & 2 about here]

4.2. Mishkin test

To formally test the treatment effect of our intervention, we apply the Mishkin test outlined by Sloan (1996). This approach is a system of two equations:

$$Earnings_{t+1} = \gamma_0 + \gamma_1 Accruals_t + \gamma_2 Cash Flow_t + v_{t+1} \quad (1)$$

$$Abnormal Return_{t+1} = \beta(Earnings_{t+1} - \gamma_0 - \gamma_1^* Accruals_t - \gamma_2^* Cash Flow_t) + e_{t+1} \quad (2)$$

Equation (1) is the earnings forecasting equation (Equation [10] in Sloan, 1996). It uses a firm's *Accruals* and *Cash Flow* in the current period (year t) to predict the firm's *Earnings* in the next period (year $t+1$). A low persistence of accruals relative to cash flows predicts $\gamma_1 < \gamma_2$.

Equation (2) is the rational pricing model (Equation [11] in Sloan, 1996). It relates the earnings surprises (*Earnings* in $t+1$ minus the *Earnings* predicted by accruals and cash flows in t) to the firm's future abnormal returns. If the market is efficient, then $\gamma_1^* = \gamma_1$ and $\gamma_2^* = \gamma_2$. That is, accruals and cash flows are priced precisely according to their contribution to earnings. Furthermore, if accruals exhibit less earnings persistence than cash flows ($\gamma_1 < \gamma_2$), market efficiency also dictates that $\gamma_1^* < \gamma_2^*$. Alternatively, if stock prices behave as if investors cannot distinguish between the two components of earnings, the coefficients on the two components are equal (i.e., $\gamma_1^* = \gamma_2^*$).

To estimate the system, we measure *Accruals* _{t} and *Cash Flow* _{t} based on the financial statements announced during our experiment period. *Earnings* _{$t+1$} is the earnings in the year following our experiment. *Abnormal Return* _{$t+1$} is *CAR*(11,251), a one-year, cumulative value of abnormal returns following the experiment. The summary statistics for these variables can be found in Appendix 2. We categorize the financial statement variables based on decile rankings and estimate the system for each of our sample groups. The results are reported in Table 3.

[Insert Table 3 about here]

Panel A reports the estimates for the Chinese market. In group S (Columns [1] and [2]), γ_2 (0.642) is higher than γ_1 (0.428), indicating lower earnings persistence of accruals relative to cash flows. However, γ_2^* (1.108) is lower than γ_1^* (1.293), suggesting that the stock prices do not rationally anticipate the lower (higher) persistence of earnings performance attributable to the accrual (cash flow) components of earnings. Notably, $\gamma_1^* > \gamma_1$ and $\gamma_2^* > \gamma_2$. This suggests an overpricing of accruals and cash flows whereby the future earnings level implied by the current

stock prices surpasses the level implied by the earnings persistence of accruals and cash flows. The market efficiency test yields a likelihood ratio statistic of 5.115 (P -value: 0.077). The null hypothesis of market efficiency ($\gamma_1^* = \gamma_1$ and $\gamma_2^* = \gamma_2$) is thus rejected. Group C, detailed in Columns (3) and (4), mirrors these findings, further supporting evidence of accrual mispricing.

In the treatment group T₁, as illustrated in Columns (5) and (6), γ_2 (0.573) is greater than γ_1 (0.399), supporting the lower earnings persistence of accruals relative to cash flows. Furthermore, γ_2^* drops to 0.578, which is almost identical to γ_2 , implying the accurate pricing of cash flows. While γ_1^* (0.870) remains higher than γ_1 , this gap narrows, in contrast to the patterns observed in groups S and C. This hints at a reduction in accrual mispricing. With a likelihood ratio of 3.712 (P -value: 0.156), the data do not reject the null hypothesis of market efficiency. This provides evidence, albeit weak, of a decrease in accrual mispricing within group T₁.

For the treatment group T₂, as presented in Columns (7) and (8), γ_2 (0.615) exceeds γ_1 (0.425), and γ_2^* (0.556) is also greater than γ_1^* (0.478), suggesting that stock prices rationally anticipate the earnings persistence associated with accrual and cash flow components. Notably, γ_1^* closely mirrors γ_1 , and γ_2^* does not surpass γ_2 . The likelihood ratio is 0.256 (P -value: 0.879), thus not rejecting the null hypothesis of market efficiency. This offers compelling evidence that accruals mispricing is reduced in group T₂.

The estimates for the US market are reported in Panel B. In groups S and C, γ_2 consistently exceeds γ_1 . However, γ_2^* is less than γ_1^* in group S, and γ_2^* is close to γ_1^* in group C. In both groups, $\gamma_1^* > \gamma_1$ and $\gamma_2^* > \gamma_2$, and the likelihood ratio is close to 5 (P -value is below 0.1). The results support the existence of accrual mispricing in groups S and C.

For the treatment group T₁, γ_2 (0.478) is greater than γ_1 (0.238). In comparison to the patterns in groups S and C, both γ_2^* (0.860) and γ_1^* (0.974) are largely reduced. The disparities between these and their respective coefficients in the forecasting equation are also diminished. With a likelihood ratio of 1.345 (P -value: 0.510), the null hypothesis of market efficiency is thus not rejected. This indicates a reduction in accrual mispricing within group T₁.

In the treatment group T₂, γ_2 (0.344) surpasses γ_1 (0.229), and γ_2^* (1.046) is also higher than γ_1^* (0.820). Notably, while the γ_1 values are similar in T₁ and T₂, γ_1^* in T₂ is reduced relative to γ_1^* in T₁. This suggests a more pronounced mitigation of accrual mispricing in T₂. Additionally, in comparison to group C, the differences between γ_1^* and γ_1 and between γ_2^* and γ_2 are lowered.

The likelihood ratio is 2.085 (P -value: 0.353), thus not rejecting the null hypothesis of market efficiency. These results suggest that accruals mispricing is mitigated in this treatment group.

In summary, the Mishkin test confirms the presence of accrual mispricing in both the Chinese and US markets. This mispricing is reduced in the treatment groups across both nations. Notably, the mitigating effect seems more potent in group T_2 than in group T_1 for both markets.

4.3. Education treatment and accrual pricing at EAs

The Mishkin test demonstrates that our accrual knowledge treatment reduces accrual mispricing. Accrual mispricing comprises two components: (i) accruals are overpriced at EAs and (ii) the prices reverse in the post-EA period. To substantiate our findings, we delve into both of these components. We assess the first component in this subsection. The second component is explored in the subsequent subsection.

To investigate whether our treatment mitigates accrual overpricing at EAs, we build upon previous studies that regress EA abnormal returns on accruals (see Balsam et al., 2002; Battalio et al., 2012; Louis et al., 2008; Miao et al., 2016). Specifically, we apply the following model:

$$CAR(0,1)_{i,m} = \beta_1 + \beta_2 Treat_i \times Accruals_{i,m}(Accruals_{UD,i,m}) + \beta_3 Treat_i + \beta_4 Accruals_{i,m}(Accruals_{UD,i,m}) + \beta_5 SUE_{i,m} + \mathbf{J} + \mathbf{M} + \varepsilon_{i,m}, \quad (3)$$

where $CAR_{i,m}$ is the two-day EA return for firm i , which announced its 2019 earnings in month m of 2020. $Accruals_{i,m}$ ($Accruals_{UD,i,m}$) is the normal accruals (unexpected discretionary accruals), constructed based on the financial statement information disclosed during the EAs.

$Treat_i$ is our treatment indicator. It equals one if firm i is in group T_1 or T_2 , and zero if it is in group C. We use $Treat_i$ to gauge the average treatment effect. To distinguish between the conceptual and methodological treatment effects, we define $T_{1,i}$ and $T_{2,i}$, respectively. $T_{1,i}$ ($T_{2,i}$) equals one if firm i is in group T_1 (T_2) and zero if it is in group C. For robustness, we alternatively use the group S as the control group, in which case the treatment indicators are set to one for firms in groups T_1 or T_2 and zero for firms in group S.

We include $SUE_{i,t}$ to control for the component of accruals that is related to earnings news. In theory, the accruals should not correlate with the announcement returns after controlling for the earnings news when accrual mispricing is absent. We also control for industry fixed effects ($\mathbf{J}$). Lastly, due to significant stock price changes in early 2020 for the COVID-19 pandemic, we further control for the fixed effects of EA months ($\mathbf{M}$).

We estimate Equation (3) by running regressions with weighted least squares. Recent studies of stock price anomalies emphasize the economic importance and statistical reliability of value-weighted returns and cross-sectional regressions with weighted least squares (Green et al., 2011; Hou et al., 2015, 2020). In line with these studies, our regressions use weighted least squares, taking the market value of equity at the end of 2019 as the weights. Unless stated otherwise, all of the estimations of Equation (3) in this paper are conducted on a value-weighted basis.

We conduct a placebo test. Specifically, we estimate the equation using an annual panel spanning from fiscal year 2013 to fiscal year 2018. The results can be found in Internet Appendix 7. Consistent with the existence of accrual mispricing, the coefficients on *Accruals* and *Accruals_{UD}* are significant and positive in both of the markets. Importantly, the coefficient on the interaction term between *Treat* and *Accruals* (*Accruals_{UD}*) is indistinctive from zero. The results suggest that the treatment effects, if any, are falsifiable.

Table 4 provides the estimates for Equation (3), while Appendix 2 offers a summary of the statistics for the variables used in this analysis. We find that our education treatment results in a discount on the pricing of accruals in both the Chinese and US markets.

[Insert Table 4 about here]

Panel A presents the average treatment effects. Columns (1) and (2) show the results for the Chinese market. The coefficient on *Accruals* is 0.02, significant at the 1% level. The coefficient on *Treat* $\times$ *Accruals* is -0.03, significant at the 1% level. In other words, a one-standard-deviation increase in *Accruals* for treatment stocks is associated with a 0.3% (0.09×0.03) decline in *CAR*(0,1). This suggests that when investors receive the education treatment, they no longer take a positive view of a high *Accruals* value. Similar results are found in Column (2), where the accrual measure is *Accruals_{UD}*. The coefficients on *Treat* $\times$ *Accruals_{UD}* and *Accruals_{UD}* have similar magnitudes but opposite signs, suggesting that investors receiving the treatment no longer take a positive view of high *Accruals_{UD}* values.

Columns (3) and (4) report the results for the US market. The coefficient on *Treat* $\times$ *Accruals* is -0.11, significant at the 5% level. This implies that relative to control stocks, there is a reduction of 0.8% (0.07×0.11) in *CAR*(0,1) for a one-standard-deviation increase in *Accruals* for the treatment stocks. The treatment also introduces a discount on the pricing of *Accruals_{UD}*. A one-standard-deviation increase in *Accruals_{UD}* for the treatment stocks is associated with 0.7% (0.13×0.05) lower announcement returns relative to the control stocks.

It is worth mentioning that the coefficients on *Treat* alone are significant and negative in Columns (1) to (3). This finding implies that our education treatment may allow investors to assess firms' earnings quality beyond the measures of *Accruals* and *Accruals_{UD}*.

Panels B and C report the treatment effects for T_1 and T_2 , respectively. The results associated with T_2 are stronger than those for T_1 . This suggests that the methodological knowledge treatment offers incremental benefits as compared to the conceptual knowledge treatment.

Specifically, in the Chinese market, the coefficients on both $T_1 \times \text{Accruals}$ and $T_2 \times \text{Accruals}$ are negative. However, the significance level of the latter ($t\text{-value} = 3.34$) surpasses that of the former ($t\text{-value} = 2.66$). The coefficient on $T_1 \times \text{Accruals}_{UD}$ stands at -0.04, significant at the 5% level, while the coefficient on $T_2 \times \text{Accruals}_{UD}$ is -0.07, significant at the 1% level. In the US market, the coefficient on $T_2 \times \text{Accruals}$ (and $T_2 \times \text{Accruals}_{UD}$) is negative and more significant than that on $T_1 \times \text{Accruals}$ ($T_1 \times \text{Accruals}_{UD}$). These results align with the observations presented in Figures 1 and 2, where group T_2 stock prices are notably more suppressed than those in T_1 .

We re-estimate the equation using samples from groups T_1 , T_2 , and S. We re-code the treatment indicator, assigning a value of one for firms in either group T_1 or T_2 and zero for firms in group S. The results are available in Internet Appendix 8. They follow the same pattern except that the difference between the treatment effects of T_1 and T_2 are more salient in the Chinese market.

4.4. Education treatment and accrual pricing in subsequent periods

The above evidence indicates that accrual education mitigates investors' overreaction to accruals. If this is indeed the case, the treatment stocks should be less likely to experience a price reversal in the subsequent period than the control stocks. We test this prediction in this subsection, applying both portfolio and regression analyses.

4.4.1. Portfolio analysis

We form portfolios labeled "Low" and "High" to reflect the associated accruals within each of our sample groups. The Low and High accrual portfolios consist of stocks with *Accruals_{UD}* in the bottom and top deciles of each group, respectively. Within each group, we form a hedge portfolio by purchasing stocks in the Low category and shorting those in the High category. We then compute the Size/BM adjusted returns of these portfolios from day 11 to day 251. To achieve this, we first determine the daily abnormal returns for each stock by subtracting the returns of the

corresponding 5×5 Size/BM matched portfolio. Next, we calculate a portfolio's daily abnormal returns by computing the value-weighted average of the abnormal returns of the stocks within that portfolio. Finally, the Size/BM adjusted return is the average of a portfolio's abnormal returns spanning from day 11 to day 251.

Additionally, we compute the portfolios' alpha using the Fama–French three-factor model. Specifically, we first determine a portfolio's daily returns by calculating the value-weighted average of the raw returns of the stocks within that portfolio. The alpha is the constant derived from regressing a portfolio's daily excess returns (returns minus the risk-free rate) on the Fama–French three factors from day 11 to day 251. Table 5 presents the results.

[Insert Table 5 about here]

Panel A presents the results for the Chinese market. The Size/BM adjusted returns are shown in Panel A1. In group S, as shown in Column (1), the daily average adjusted returns for the Low and High portfolios are 0.055% and -0.017%, respectively. This aligns with the idea of accrual mispricing, suggesting that higher accruals correlate with reduced returns in subsequent periods. The hedge portfolio yields an average return of -0.072% (equivalent to an annualized return of 18%), which is significant at the 10% level. A similar trend is observed in group C, as presented in Column (2). In group T₁, depicted in Column (3), the return for the High portfolio turns positive and the hedge portfolio's return is no longer statistically significant, suggesting a reduction in accrual mispricing. In group T₂, as seen in Column (4), the returns for both the High and hedge portfolios are positive and nonsignificant, indicating the absence of accrual mispricing in group T₂. Panel A2 presents the alpha based Fama–French three-factor model. The pattern is the same.

The results for the US market are presented in Panel B. As shown in Panel B1, the Size/BM adjusted returns for the Low and High portfolio in group S stand at 0.044% and -0.031%, respectively. The hedge portfolio return is -0.075%, significant at the 1% level. Group C's hedge portfolio return is negative at 0.196%, significant at the 1% level. In group T₁, the High portfolio's return is positive yet nonsignificant at 0.018%, while the hedge portfolio yields a positive return of 0.041%, significant at the 10% level. For group T₂, the returns for the High portfolio and the hedge portfolio are 0.051% and 0.052%, respectively, both of which are statistically significant. Panel B2 show that the alphas of the hedge portfolio for groups S and C are significant and negative but are nonsignificant for the two treatment groups.

4.4.2. Regression analysis

Next, we run a regression analysis. Specifically, we estimate Equation (3) by replacing the dependent variable with the one-year abnormal return after the experiment ($CAR(11,251)$). The results are reported in Table 6. As expected, high accruals correlate with lower future returns, but this negative relationship is mitigated by the education treatment.

Pane A presents the average treatment effects. For the Chinese market, as depicted in Column (1), a one-standard-deviation increase in *Accruals* is associated with an approximately 3.2% (0.09×0.35) reduction in $CAR(11,251)$ for the control stocks. The decline is reduced to only 0.6% ($0.09 \times (0.35 - 0.28)$) for the treatment stocks. A one-standard-deviation increase in *Accruals_{UD}* corresponds to an 8.2% (0.13×0.63) drop in $CAR(11, 251)$ for the control stocks, as shown in Column (2). This negative relationship diminishes for the treatment stocks, given that the treatment increases the future stock return by around 10.9% (0.13×0.84).

For the US market, a one-standard-deviation increase in *Accruals* for the control stocks is associated with a 6.1% (0.07×0.87) decline in $CAR(11, 251)$, as shown in Column (3). This reduction is lowered to 0.6% ($0.07 \times (0.87 - 0.78)$) for the treatment stocks. Column (4) reveals that a one-standard-deviation increase in *Accruals_{UD}* leads to an approximately 2.9% (0.13×0.22) decline in $CAR(11, 250)$ for the control stocks. The reduction disappears for the treatment stocks, as the education treatment improves the return by 5.3% (0.13×0.41).

We observe that the coefficients on *Treat* are significant and negative across all of the columns. This suggests that our educational program may have enduring effects on the pricing of accruals announced in subsequent quarters. Supporting this perspective, as we detail in Section 6, the treatment stocks with high accruals continue to experience a pricing discount during EAs as compared to control stocks, even a year after the experiment.

The treatment effects of T_1 and T_2 are reported in Panels B and C, respectively. Consistent with previous findings, the treatment effect of T_2 is stronger than that of T_1 . Specifically, except for $T_1 \times Accruals_{UD}$ in the Chinese market, the coefficients on the interaction terms associated with T_1 are nonsignificant. In contrast, all of the coefficients on the interaction terms associated with T_2 are significant and have expected signs. For example, the coefficients on $T_2 \times Accruals$ stand at 0.38 in the Chinese and 1.28 in the US markets, both significant at the 1% level. These results underscore the importance of supplementing conceptual knowledge with methodological insights to enhance investors' learning.

The results of using group S as the control are reported in Internet Appendix 9. The pattern remains unchanged in the Chinese market. In the US market, the coefficients on the interaction between *Treat* and accruals are positive but not statistically significant. When breaking down the average treatment into the two sub-treatments, we observe a more pronounced treatment effect for T_2 than for T_1 . These findings confirm the dominant treatment effect of T_2 .

Overall, our results indicate that our education treatment mitigates accrual overpricing and reduces price reversals in subsequent periods in both the Chinese and US markets. Importantly, the effect is amplified when methodological knowledge is provided alongside conceptual knowledge.

[Insert Table 6 about here]

5. Heterogeneity effects

Our findings suggest that our education treatments significantly mitigate accrual mispricing. In this section, we explore the heterogeneous effects of our results.

5.1. Engagement with educational materials

The effectiveness of education hinges on the degree to which individuals engage with the educational content. If our experiment truly has influenced the pricing of accruals through the educational process, we would expect to find a more pronounced treatment effect when investors more actively engage with our materials.

To test this, we introduce the variable *Reads* to gauge the volume of investors' engagement with our social media posts. *Reads* is set to one if the total number of posts read across all of the social media platforms for a stock exceeds the bottom tercile, and zero otherwise. Additionally, we use the variable *CSLS* to measure the depth of investors' interactions with our posts. *CSLS* is set to one if the combined number of comments or replies on our posts for a specific stock across all of the platforms exceeds the bottom tercile, or if the posts are "retweeted/shared," "liked," or "saved" on any of the platforms, and zero otherwise. We also introduce a variable, *WebView*, to track the number of visits to webpages containing our full educational content. *WebView* is set to one if the number of clicks on both type 1 and type 2 link pages for a stock surpasses the bottom tercile among the treatment stocks, and zero otherwise. With these variables in place, we then incorporate *Reads*, *CSLS*, and *WebView* into Equation (3) and re-run the regressions.

The results are reported in Table 7, where the dependent variable is $CAR(0,1)$. To save space, we only report the estimates for the unexpected discretionary accruals ($Accruals_{UD}$). In the Chinese market, the coefficients on $Treat \times Accruals_{UD} \times Reads$, $Treat \times Accruals_{UD} \times CSLS$, and $Treat \times Accruals_{UD} \times WebView$ are negative and significant at the 1% level (see Panel A). This suggests a more pronounced discount on the pricing of accruals for the treatment stocks when investors engage more intensively with our educational materials.²³

In the US market, the coefficient on $Treat \times Accruals_{UD} \times WebView$ is also negative and significant. The coefficient on $Treat \times Accruals_{UD} \times Reads$ and $Treat \times Accruals_{UD} \times CSLS$ is negative but statistically nonsignificant. The subdued results associated with *Reads* and *CSLS* align with the limited engagement features available on US social media platforms as compared to those available in China. For instance, StockTwits lacks a “read” function, and neither Twitter nor StockTwits offers a “save” option. This reduced functionality impacts the variability and, consequently, undermines the measurement accuracy of *Reads* and *CSLS*.

Turning to the results for the treatment indicators T_1 and T_2 (see Panels B and C), we find that *Reads*, *CSLS*, and *WebView* amplify the treatment effects of both T_1 and T_2 in the Chinese market. In the US market, *Reads* increases the treatment effects of T_2 , while *CSLS* enhances the treatment effect of T_1 and the standalone effect of T_2 .

We alternatively use group S as the control, and the results are available in Internet Appendix 10. A notable enhancement of the results is evident. For instance, in the Chinese market, the treatment effect of T_1 , despite initially not being statistically significant as indicated in Internet Appendix 8, achieves significance when factoring in investors with heightened learning engagement. Similarly, in the US market, the average treatment effect (denoted as *Treat*) emerges as statistically significant with the anticipated signs when considering investors who engage more deeply with our educational content. Collectively, these findings suggest that the efficacy of the educational intervention is magnified when investors invest more effort into our provided materials.

[Insert Table 7 about here]

5.2. Retail investors

²³ The interaction terms $Treat \times Webview$ and $Accruals_{UD} \times Webview$ are omitted to avoid multicollinearity because *WebView* indicates a subgroup of the treatment stocks and takes the value of zero in all of the control stocks (stocks in group C do not have link pages).

Retail investors are the primary users of social media. Compared to institutional investors, They have less accounting and financial knowledge and a limited understanding of the implications and applications of accruals (Balsam et al., 2002). As such, we expect our education program to be particularly beneficial for stocks dominated by retail investors. We measure retail investor intensity as 100% minus the percentage of shares held by institutional investors and introduce a dummy variable, RI , which is assigned a value of one when retail investor intensity exceeds the sample's bottom tercile, and zero otherwise. With RI incorporated, we re-estimate Equation (3).

The results are reported in Table 8. In the Chinese market, the coefficient on $Treat \times Accruals_{UD} \times RI$ is significant and negative, suggesting that our accrual education program is particularly helpful for retail investors in China. The coefficient on the interaction term is also negative in the US market. However, it lacks statistical significance.

When examining the separate treatment indicators, T_1 and T_2 , the coefficient for the interaction term of T_1 is significant and negative in the Chinese market. However, the coefficient for T_2 is not statistically significant. This suggests that providing solely conceptual knowledge is more effective for retail investors in China than offering both conceptual and methodological knowledge. In the US market, the coefficients for the interaction terms of both T_1 and T_2 remain nonsignificant.

We alternatively use group S as the control group and present the results in Internet Appendix 11. The effects associated with T_1 and T_2 remain consistent in the Chinese market. Interestingly, the coefficients on the interaction term of all three treatment indicators in the US market become significant and negative. This shift, compared to the analysis when the control group is C, may indicate that retail investors in the US are less likely to overact to firms' accruals if they are reminded to focus on the firms' EAs. This is logical, as retail investors are less susceptible to bias if they are guided to focus on a firm's earnings and underlying fundamentals (Lawrence et al., 2018).

The disparities between the two countries align with the idea that Chinese investors are generally less sophisticated than their US counterparts. Chinese retail investors are often hindered by greater cognitive limitations and have a lower learning capacity (Chen et al., 2004). Previous studies show that adding more data or details can make it more difficult for investors who have limited information-processing capacity to extract information (information overload problem) (Lu, 2022). Therefore, focusing on the basic conceptual understanding of accruals is more effective

for Chinese retail investors than considering a mix of both conceptual and methodological knowledge (i.e., more is less). In contrast, US retail investors are relatively sophisticated, making it advantageous for them to receive both types of knowledge.

[Insert Table 8 about here]

6. Alternative explanations and robustness

In this section, we conduct additional analyses to test for alternative explanations.²⁴

6.1. Salience of the treatment

One may have the concern that investors are influenced more by the salience of our intervention than by the educational content itself. Although we adopt a DID approach to difference out the salience effect by comparing treatment stocks with control stocks that receive a “placebo” treatment, the salience effect may bias our findings if it interweaves with our education effects. To mitigate this concern, we run a regression analysis by comparing accrual pricing between the control group C and the comparison group S that receives no intervention from us.

We repeat the baseline analysis, as shown in Panel A of Table 4, this time focusing on the sample comprising groups C and S. *Treat* is coded one for stocks in C and zero for stocks in S. The results can be found in Internet Appendix 12. The coefficients on both $Treat \times Accruals$ and $Treat \times Accruals_{UD}$ are nonsignificant. Interestingly, the coefficients are positive in the Chinese market, which runs counter to the expected direction of the education effect. This reinforces our belief that the observed educational effect is not merely a byproduct of the treatment’s salience.

6.2. Attention to accruals

One may also raise the concern that our findings are due not to the education effect but to increased attention to accruals. First, our results may be attributable to investors’ reactions to historical accruals that we present on our website. Second, our experimental exercise may prompt investors who are already familiar with accruals to apply their pre-existing knowledge.

To address the first possibility, we reconfigure Figures 1 and 2, focusing on the historical accrual data incorporated into the spreadsheet tool on our webpage for group T₂. If our findings

²⁴ The results of this section, if unreported, are available upon request.

are influenced by investors' responses to accruals they previously encountered, we would expect to observe diverse market reactions based on the accrual rankings displayed on the website. However, our analysis reveals that the price movements are largely consistent across the high and low accrual rankings.

To rule out the second possibility, we leverage our experimental design that emphasizes positive accruals. We educate investors about the implications of high accruals, aiming to simplify our materials for easier comprehension. If our exercise merely reminds investors to apply pre-existing knowledge, the treatment effect should be evident for both positive and negative accruals, with the investors' interpreting positive (negative) accruals as bad (good) news. However, if our exercise truly imparts an education effect, the treatment should be relevant only for positive accruals. To test this, we re-run our regression analysis, focusing on sub-samples of firms with either positive (the top decile) or negative (the bottom decile) accruals.

The results are reported in Internet Appendix 13. Panel A reports the results for the firms with positive accruals. We find that the treatment effects are highly significant and the magnitude of the interaction between treatment indicators and accruals increases as compared to what is shown in Panel A of Table 4. For the negative accruals, as reported in Panel B, the treatment effects are statistically nonsignificant. The results thus do not support the view that the treatment only causes investors to recall their pre-existing accrual knowledge.

6.3. Investor deterrent effect

Another concern is that our education may deter investors due to its cautionary tone, which could be perceived as fear-inducing. Such a tone may intimidate investors, leading them to avoid stocks with high accruals. This could result in a subdued short-term market reaction and a reduced long-term reversal. To address this alternative explanation, we assess the persistence of the treatment effect one year post-experiment, specifically in 2021. The underlying rationale is straightforward: if our findings are primarily attributable to a deterrent effect, they should dissipate during the 2021 EAs, when the experiment and any fear-inducing messages are absent. Conversely, if the observed effects stem from the genuine education impact, they should persist beyond the experimental phase, because effective education tends to have a lasting influence on individuals.

Specifically, we re-estimate Equation (3) to investigate the impact of the education treatment on the markets' short-term reaction to firms' 2020 accruals (announced in 2021). The results are

reported in Internet Appendix 14. We find that high accruals in 2020 for treatment stocks continue to experience a significant pricing discount relative to the control stocks in both markets. This significance persists when we partition the treatment variable *Treat* into T_1 and T_2 .

6.4. Spillover effect

It is possible that some investors own or trade both treatment and control stocks, which leads to the transmission of the treatment effect to the control stocks. We argue that this spillover effect, if it exists, should have a limited impact on our treatment effects. First, the spillover effect may reduce the magnitude of our estimate without biasing the direction of the estimation.

Second, prior studies show that households own relatively few individual stocks (Campbell, 2006). For example, the Survey of Consumer Finances (SCF) shows that the median number of stocks held by households is only two. Supporting this, a substantial number of studies document that individual investors tend to make concentrated holdings in certain stocks because of local bias or familiarity (e.g., they buy stocks of hometown firms or their employer firms) (Goetzmann and Kumar, 2008; Van Nieuwerburgh and Veldkamp, 2009). Notably, these studies show that underdiversification is concentrated on individuals with low financial literacy (Von Gaudecker, 2015). As a result, the spillover effect should have a limited effect on our estimation.

Nevertheless, we conduct an analysis to validate our perspective. We use stock turnover, calculated as the daily average of traded shares over total shares outstanding, as the measure of trading frequency. Stocks with high turnover exhibit frequent trading and are apt to be traded or held by various investors within a short window (Choi et al., 2002). We find that the pricing discount of the treatment firms as compared to the control firms remains significant in stocks with both high and low turnover, suggesting that the impact of the spillover effect is limited.

6.5. Impact of the COVID-19 pandemic

Our experiment was designed in 2019, but unfortunately, its execution coincided with the outbreak of the COVID-19 pandemic. This unforeseen event may affect our results. However, we believe that the impact of this event is minimal because the data released during the pandemic pertained to the financial performance of 2019, which had already been finalized. Despite this, we take several measures to mitigate this concern. First, we control for the EA months' fixed effects (M) in our regression (Equation 3). Second, when excluding firms filing EAs during the pandemic peak,

our results remain largely consistent.²⁵ Third, as shown above, the treatment effects persist one year post-experiment. By this time, the pandemic had subsided.

6.6. Other robustness tests

We conduct further analyses as robustness checks. First, we use alternative windows to measure EA returns, specifically windows from days 0 to 2 ($CAR(0,2)$), from days -5 to 5 ($CAR(-5,5)$), and from days -10 to 10 ($CAR(-10,10)$). We find similar results using these alternative windows. Second, to mitigate the concern that our results are driven by outliers, we alternatively measure *Accruals* and *Accruals_{SUD}* using decile ranks. We also find robust results using these measures. Third, we repeat our analysis based on alternative models. Specifically, we re-estimate our analysis by dropping EA month and industry fixed effects, separately. We also repeat our analysis by controlling for firm characteristics such as firm size, return on assets, cash flows, stock return, and volatility. Our results are robust to these alternative specifications.

7. Conclusion

This study investigates the influence of investor education on stock mispricing. We impart financial knowledge regarding the concept and pricing implications of accounting accruals to investors across randomized stock groups in China and the US, using social media platforms. Our aim is to determine whether this intervention mitigates accrual mispricing.

We observe the diminished mispricing of the treatment stocks compared to the control stocks in both markets. The effect is most pronounced when knowledge that enhances the conceptual understanding of accruals is paired with knowledge that aids in the quantitative application of accruals. Additionally, the impact is more potent when investors are less sophisticated and engage more with our educational materials. The education treatment effect remains robust, unaffected by treatment salience or heightened investor attention to accruals. Furthermore, the effect seems not to be attributed to investor deterrent or spillover effects.

Overall, our results suggest that investor education has a causal effect on asset pricing, signifying the power of knowledge in promoting financial market efficiency and development.

²⁵ The peak of the pandemic in China was in February and March of 2020. In the US, it was in March and April of 2020.

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Appendix 1. Sample construction and filtering

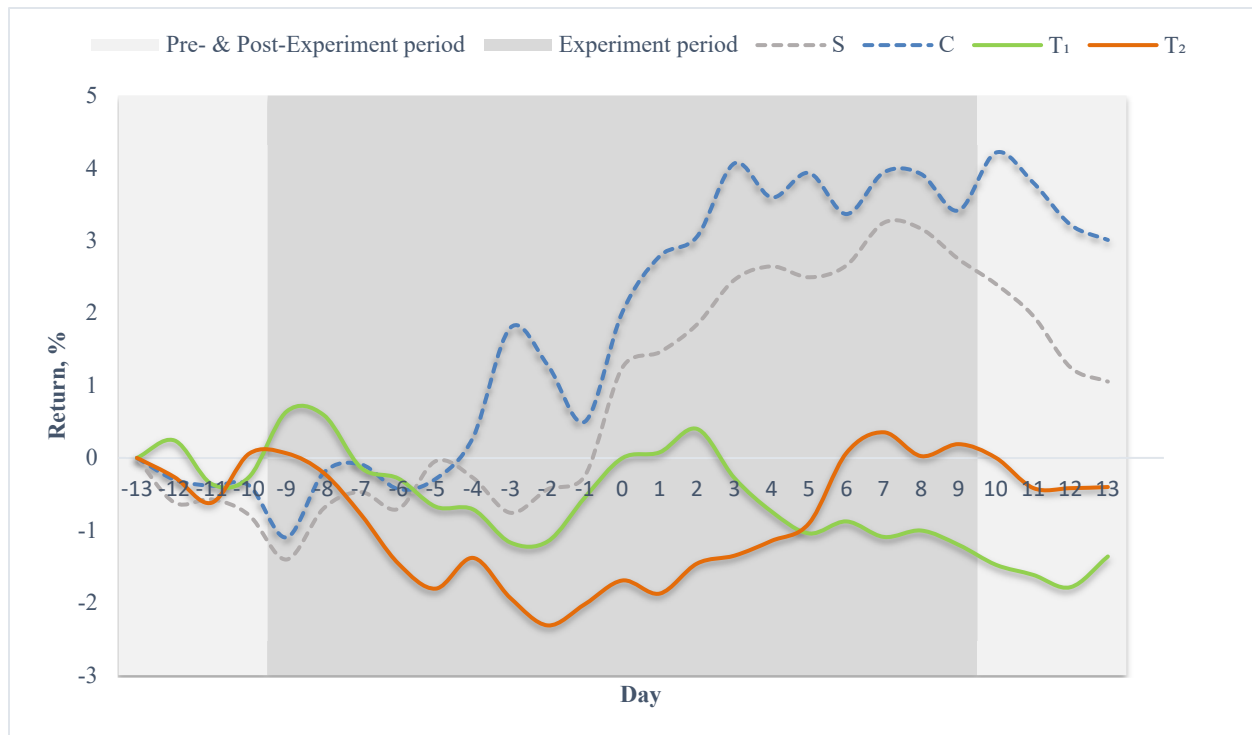
Panel A: Chinese market					
Initial number of stocks listed in China's stock markets by the end of 2018					3,689
- delete B shares					99
- delete stocks in financial and real estate industries (i.e., CSRC industry classification is "J" or "K")					317
- delete stocks with insufficient historical financial information from 2009 to 2018 ($n \leq 3$) to estimate accruals					1,062
- delete stocks with no trading information in 2018					2
Final sample for the experiment					2,308
Treatment groups	S	C	T ₁	T ₂	Total
Randomly divide the sample into four groups	577	577	577	577	2,308
- delete stocks that were delisted or did not file annual reports in 2020	2	1	3	3	9
- delete stocks that were suspended for trading in 2020	2	3	5	5	15
Final sample for analysis	573	573	569	569	2,284
Panel B: US market					
The initial number of stocks listed in US stock markets by the end of 2018 (included in CRSP/Compustat merged dataset)					5,320
- delete inactive stocks (i.e., COSTAT = "A")					127
- delete stocks in the financial and real estate industries (i.e., SIC 6000–6999)					1,508
- delete stocks with insufficient historical financial information from 2009 to 2018 ($n \leq 3$) to estimate accruals					1,093
- delete stocks with no trading information in 2018					45
Final sample for the experiment					2,547
Treatment groups	S	C	T ₁	T ₂	Total
Randomly assign the sample into four groups	636	637	637	637	2,547
- delete stocks that were delisted or did not file annual reports in 2020	25	33	25	25	108
- delete stocks that were suspended for trading in 2020	16	10	13	13	52
Final sample for analysis	595	594	599	599	2,387

Appendix 2. Variables summary statistics

Panel A: Chinese market						
Variable	(1) N	(2) Mean	(3) STD	(5) P25	(6) P50	(7) P75
The Mishkin test:						
<i>Abnormal Return_{t+1}</i>	2,284	-0.16	0.40	-0.40	-0.24	-0.00
<i>Earnings_{t+1}</i>	2,284	0.60	0.28	0.40	0.60	0.80
<i>Accruals_t</i>	2,284	0.60	0.28	0.40	0.60	0.80
<i>Cash Flow_t</i>	2,284	0.60	0.28	0.40	0.60	0.80
Regression analysis of EAs:						
<i>CAR(0,1)</i>	2,284	-0.01	0.05	-0.03	-0.01	0.02
<i>CAR(11,251)</i>	2,284	-0.16	0.40	-0.40	-0.24	-0.00
<i>Accruals</i>	2,284	-0.02	0.09	-0.07	-0.02	0.03
<i>Accruals_{UD}</i>	2,284	0.00	0.13	-0.07	-0.00	0.07
<i>SUE</i>	2,284	-0.00	0.01	-0.00	0.00	0.00
<i>Treat</i> (C is the control)	1,711	0.67	0.47	0.00	1.00	1.00
<i>T₁</i> (C is the control)	1,142	0.50	0.50	0.00	0.00	1.00
<i>T₂</i> (C is the control)	1,142	0.50	0.50	0.00	0.00	1.00
<i>Treat</i> (S is the control)	1,711	0.67	0.47	0.00	1.00	1.00
<i>T₁</i> (S is the control)	1,142	0.50	0.50	0.00	0.00	1.00
<i>T₂</i> (S is the control)	1,142	0.50	0.50	0.00	0.00	1.00
<i>Reads</i>	2,284	0.56	0.50	0.00	1.00	1.00
<i>CSLS</i>	2,284	0.47	0.50	0.00	0.00	1.00
<i>WebView</i>	2,284	0.07	0.26	0.00	0.00	0.00
<i>RI</i>	2,284	0.67	0.47	0.00	1.00	1.00
Panel B: US market						
Variable	(1) N	(2) Mean	(3) STD	(5) P25	(6) P50	(7) P75
The Mishkin test:						
<i>Abnormal Return_{t+1}</i>	2,268	0.03	0.56	-0.29	-0.04	0.25
<i>Earnings_{t+1}</i>	2,268	0.61	0.29	0.40	0.60	0.90
<i>Accruals_t</i>	2,268	0.55	0.29	0.30	0.60	0.80
<i>Cash Flow_t</i>	2,268	0.52	0.29	0.30	0.50	0.80
Regression analysis of EAs:						
<i>CAR(0,1)</i>	2,387	-0.02	0.05	-0.06	-0.01	0.01
<i>CAR(11,251)</i>	2,377	0.02	0.56	-0.30	-0.05	0.25
<i>Accruals</i>	2,387	-0.06	0.07	-0.10	-0.06	-0.03
<i>Accruals_{UD}</i>	2,387	0.02	0.13	-0.04	0.01	0.07
<i>SUE</i>	2,387	0.02	0.23	0.00	0.00	0.00
<i>Treat</i> (C is the control)	1,792	0.67	0.47	0.00	1.00	1.00
<i>T₁</i> (C is the control)	1,193	0.50	0.50	0.00	1.00	1.00
<i>T₂</i> (C is the control)	1,193	0.50	0.50	0.00	1.00	1.00
<i>Treat</i> (S is the control)	1,793	0.67	0.47	0.00	1.00	1.00
<i>T₁</i> (S is the control)	1,194	0.50	0.50	0.00	1.00	1.00
<i>T₂</i> (S is the control)	1,194	0.50	0.50	0.00	1.00	1.00
<i>Reads</i>	2,387	0.44	0.50	0.00	0.00	1.00
<i>CSLS</i>	2,387	0.44	0.50	0.00	0.00	1.00
<i>WebView</i>	2,387	0.17	0.37	0.00	0.00	0.00
<i>RI</i>	2,387	0.75	0.43	1.00	1.00	1.00

Figure 1. Cumulative abnormal returns around the experiment for stocks with high accruals in the Chinese stock market

Panel A: From day -13 to day 13



Panel B: From day -13 to day 251

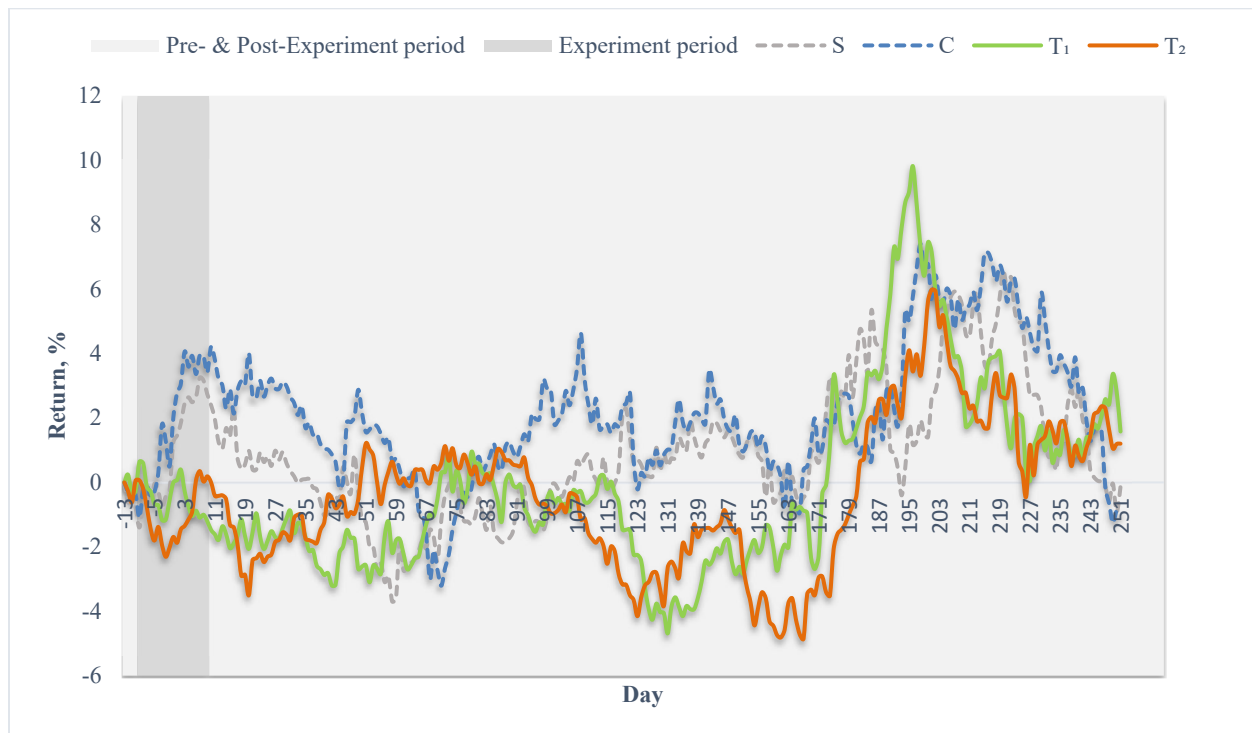
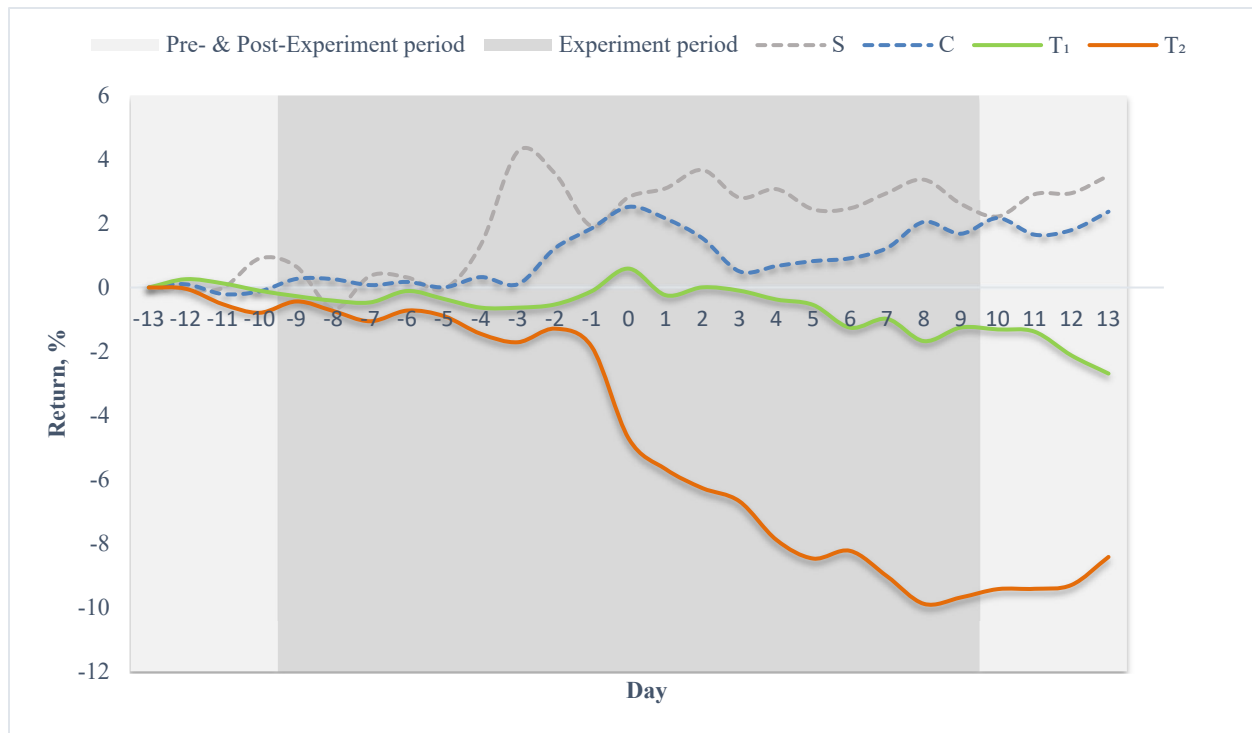


Figure 2. Cumulative abnormal returns around the experiment for stocks with high accruals in the US stock market

Panel A: From day -13 to day 13



Panel B: From day -13 to day 251

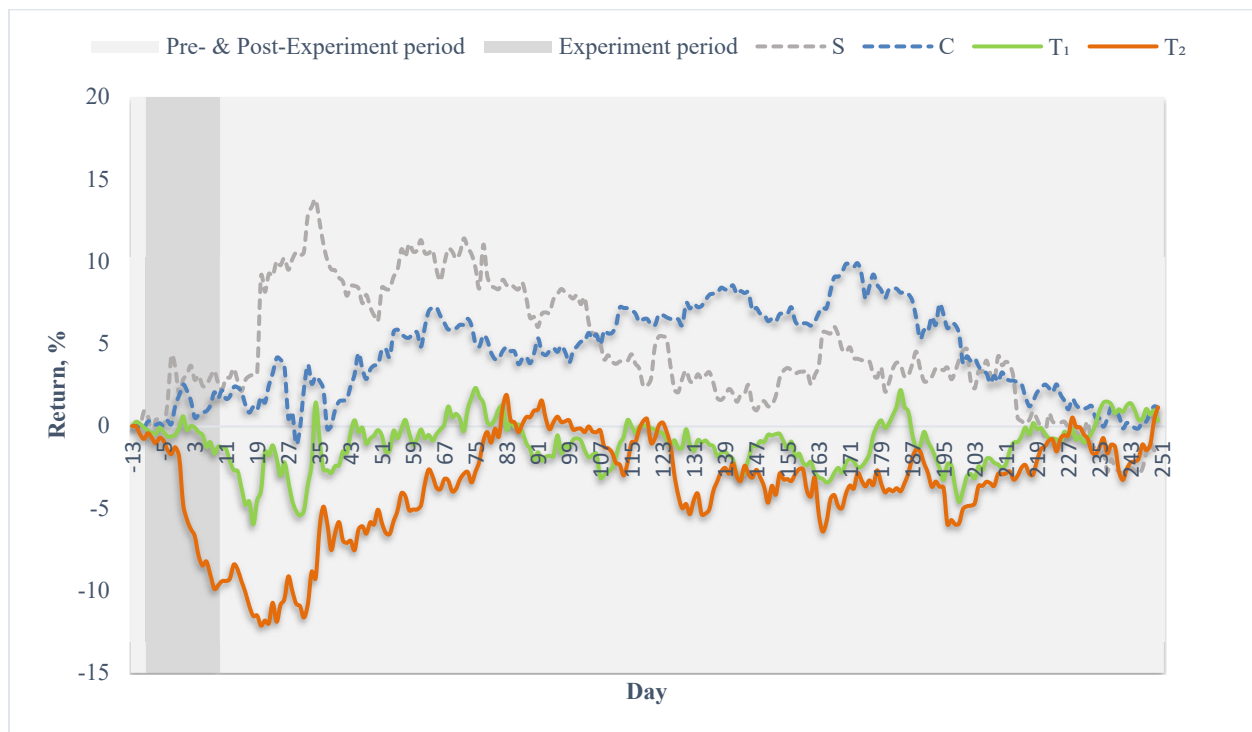


Table 1. Comparing ex-ante firm characteristics between control and treatment firms

Panel A: Chinese market										
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	S	C	T ₁	T ₂	T ₁ - C		T ₂ - C		T ₂ - T ₁	
	Ave.				Dif.	P-value	Dif.	P-value	Dif.	P-value
<i>MV</i>	15.63	15.69	15.63	15.68	-0.06	(0.35)	-0.01	(0.85)	0.05	(0.44)
<i>Leverage</i>	0.46	0.47	0.45	0.45	-0.01	(0.30)	-0.02	(0.22)	-0.00	(0.84)
<i>MB</i>	2.83	2.83	2.88	2.86	0.05	(0.80)	0.03	(0.90)	-0.03	(0.89)
<i>ROA</i>	-0.00	0.00	-0.00	0.00	-0.01	(0.46)	-0.00	(0.68)	0.00	(0.76)
<i>CashFlows</i>	0.05	0.05	0.05	0.05	-0.00	(0.73)	0.00	(1.00)	0.00	(0.73)
<i>Cash</i>	0.16	0.16	0.16	0.17	-0.00	(0.70)	0.00	(0.71)	0.01	(0.43)
<i>WorkingCapital</i>	-0.01	-0.00	0.00	-0.00	0.00	(0.52)	-0.00	(0.78)	-0.01	(0.36)
<i>Accruals</i>	-0.03	-0.03	-0.02	-0.02	0.00	(0.72)	0.00	(0.75)	-0.00	(0.96)
<i>Accruals_{DD}</i>	-0.02	-0.01	-0.02	-0.01	-0.01	(0.15)	-0.00	(0.63)	0.00	(0.37)
<i>Accruals_{KLW}</i>	-0.00	-0.00	0.00	0.00	0.00	(0.54)	0.00	(0.79)	-0.00	(0.73)

Panel B: US market										
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	S	C	T ₁	T ₂	T ₁ - C		T ₂ - C		T ₂ - T ₁	
	Ave.				Dif.	P-value	Dif.	P-value	Dif.	P-value
<i>MV</i>	20.93	20.86	20.96	21.02	0.10	(0.48)	0.16	(0.26)	0.06	(0.67)
<i>Leverage</i>	0.59	0.57	0.57	0.57	0.00	(1.00)	-0.00	(0.93)	-0.00	(0.93)
<i>MB</i>	3.24	3.14	3.29	3.21	0.15	(0.62)	0.07	(0.81)	-0.08	(0.81)
<i>ROA</i>	-0.04	-0.05	-0.04	-0.03	0.01	(0.49)	0.01	(0.28)	0.00	(0.68)
<i>CashFlows</i>	0.03	0.03	0.04	0.04	0.00	(0.67)	0.01	(0.58)	0.00	(0.88)
<i>Cash</i>	0.17	0.17	0.18	0.18	0.01	(0.30)	0.02	(0.14)	0.01	(0.66)
<i>WorkingCapital</i>	-0.00	-0.00	-0.00	-0.00	0.00	(0.35)	0.00	(0.73)	-0.00	(0.54)
<i>Accruals</i>	-0.04	-0.05	-0.04	-0.04	0.00	(0.25)	0.00	(0.24)	0.00	(0.97)
<i>Accruals_{DD}</i>	-0.00	-0.00	-0.00	-0.01	0.00	(0.87)	-0.00	(0.36)	-0.00	(0.28)
<i>Accruals_{KLW}</i>	0.01	0.01	0.01	0.01	-0.00	(0.65)	-0.00	(0.68)	0.00	(0.96)

Notes: This table reports firm characteristics from the year preceding the experiment for each sample across markets. These include the natural logarithm of market capitalization (*MV*), total liability/total assets (*Leverage*), market-to-book ratio (*MB*), return on assets (*ROA*), cash flows/total assets (*CashFlows*), cash holdings/total assets (*Cash*), working capital accruals/total assets (*WorkingCapital*), total accruals/total assets (*Accruals*), and abnormal accruals estimated based on Dechow and Dichev (2002) (*Accruals_{DD}*) and Kothari et al. (2005) (*Accruals_{KLW}*), respectively. The definitions of *WorkingCapital* and *Accruals* are consistent with the definitions in our educational materials in Internet Appendix 1.2.

Table 2. Reading volume of educational contents

Panel A: Chinese market								
A1: Total volume of all firms								
Group	(1) Posts	(2) Reads	(3) Comments /replies	(4) Retweets /reshares	(5) Likes	(6) Saves	(7) Type 1 page (Weibo)	(8) Type 2 page (Weibo)
C	18,715	9,081,608	3,526	2,732	306	97	-	-
T ₁	18,181	9,271,280	5,149	3,864	289	129	5,097	-
T ₂	17,838	8,481,201	4,385	7,108	213	166	2,982	2,817
Total	54,734	26,834,089	13,060	13,704	808	392	8,079	2,817
A2: Total volume per firm								
Group	(1) Posts	(2) Reads	(3) Comments /replies	(4) Retweet /reshares	(5) Likes	(6) Saves	(7) Type 1 page (Weibo)	(8) Type 2 page (Weibo)
C	32.49	15,767	6.12	4.74	0.53	0.17	-	-
T ₁	31.67	16,152	8.97	6.73	0.50	0.23	8.88	-
T ₂	31.08	14,776	7.64	12.38	0.37	0.29	5.20	4.91
Panel B: US market								
B1: Total volume of all firms								
Group	(1) Posts	(2) Reads (Twitter)	(3) Comments /replies	(4) Retweets /reshares	(5) Likes	(6) Saves	(7) Type 1 page (Twitter)	(8) Type 2 page (Twitter)
C	9,309	1,110,517	885	1180	1,298	-	-	-
T ₁	9,563	945,797	1,067	662	1,444	-	7,950	-
T ₂	9,557	1,341,123	738	479	1,195	-	3,270	3,810
Total	28,429	3,397,437	2,690	2,321	3,937	-	11,220	3,810
B2: Total volume per firm								
Group	(1) Posts	(2) Reads (Twitter)	(3) Comments /replies	(4) Retweets /reshares	(5) Likes	(6) Saves	(7) Type 1 page (Twitter)	(8) Type 2 page (Twitter)
C	14.61	1,743	1.39	1.85	2.04	-	-	-
T ₁	15.01	1,485	1.68	1.04	2.27	-	12.46	-
T ₂	15	2,105	1.16	0.75	1.88	-	5.13	5.98

Notes: This table reports the reading volume of the educational content. The indicator for reading volume of social media posts is the reading volume one day after each post. The indicators available vary from platform to platform. Specifically, they are reads (Guba EastMoney, Xueqiu, Weibo, and Twitter), comments (EastMoney, Xueqiu, Weibo, and Twitter), retweets or reshares (EastMoney, Xueqiu, Weibo, and Twitter), likes (Xueqiu, Weibo, Twitter, and StockTwits), reshares (Twitter and StockTwits), replies (StockTwits), and saves (Xueqiu). Web traffic data for China are sourced exclusively from Weibo, while for the US, they are sourced exclusively from Twitter. Panels A1 and B1 show the aggregate volume for all of the firms within each sample across the markets. Panels A2 and B2 detail the volume per firm within each sample across the markets.

Table 3. Mishkin test

$$Earnings_{t+1} = \gamma_0 + \gamma_1 Accruals_t + \gamma_2 Cash Flow_t + v_{t+1}$$

$$Abnormal Return_{t+1} = \beta(Earnings_{t+1} - \gamma_0 - \gamma_1^* Accruals_t - \gamma_2^* Cash Flow_t) + e_{t+1}$$

Panel A: Chinese market

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	S		C		T ₁		T ₂	
Parameter	Estimate	z-stat	Estimate	z-stat	Estimate	z-stat	Estimate	z-stat
γ_1	0.428	9.41	0.434	9.08	0.399	8.60	0.425	9.01
γ_1^*	1.293	3.40	1.011	3.84	0.870	3.02	0.478	1.92
γ_2	0.642	14.12	0.594	12.45	0.573	12.35	0.615	13.05
γ_2^*	1.108	3.42	0.666	2.82	0.578	2.18	0.556	2.23
β	0.243	3.65	0.313	4.86	0.349	4.18	0.276	4.52

Test of market efficiency: $\gamma_1 = \gamma_1^*$ and $\gamma_2 = \gamma_2^*$

Likelihood ratio	5.115	6.119	3.712	0.258
P-value	0.077	0.047	0.156	0.879

Observations	573	573	569	569
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Panel B: US market

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	S		C		T ₁		T ₂	
Parameter	Estimate	z-stat	Estimate	z-stat	Estimate	z-stat	Estimate	z-stat
γ_1	0.319	8.20	0.215	5.72	0.238	6.21	0.229	6.21
γ_1^*	1.233	2.75	1.561	2.46	0.974	1.49	0.820	1.62
γ_2	0.456	12.00	0.540	14.19	0.478	12.49	0.344	9.19
γ_2^*	1.110	2.89	1.645	2.93	0.860	1.56	1.046	1.93
β	0.029	2.89	0.022	2.55	0.017	1.79	0.018	2.10

Test of market efficiency: $\gamma_1 = \gamma_1^*$ and $\gamma_2 = \gamma_2^*$

Likelihood ratio	4.696	4.923	1.345	2.085
P-value	0.096	0.085	0.510	0.353

Observations	558	572	569	569
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Notes: This table presents the results from estimating stock price reactions to information in the accrual and cash flow components of current earnings about future earnings. The sample consists of stocks in the treatment groups T₁ and T₂, the control group C, and comparison group S. $Abnormal Return_{t+1}$ is $CAR(11,251)$, which is the buy-and-hold return from day 11 to day 251 (day 0 is the EA date for fiscal year 2019) minus the return of the matched portfolio of 5×5 size, and book-to-market portfolios in the same window. $Accruals_t$ is the change in current assets minus the change in cash holding minus the change in current liabilities plus the change in short-term debt plus depreciation, scaled by total assets in year t (fiscal year 2019). $Earnings_t$ is operating income before depreciation, scaled by total assets. $Cash Flow_t$ is the difference between $Earnings_t$ and $Accruals_t$. The variables from the financial statements are standardized based on their decile rank, scaled by ten.

Table 4. Financial education and accrual pricing in the short-term window

Panel A: Average treatment ($Treat$)				
	Chinese market		US market	
	(1)	(2)	(3)	(4)
$Treat \times Accruals$	-0.030*** (-3.47)		-0.107** (-2.29)	
$Accruals$	0.020*** (3.33)		-0.045 (-1.17)	
$Treat \times Accruals_{UD}$		-0.058*** (-4.00)		-0.052* (-1.73)
$Accruals_{UD}$		0.059*** (6.03)		-0.007 (-0.29)
$Treat$	-0.011*** (-5.53)	-0.009*** (-4.12)	-0.008** (-2.38)	-0.001 (-0.47)
SUE	0.466*** (4.99)	0.446*** (4.80)	0.021* (1.72)	0.021* (1.72)
Industry and month FEs	Yes	Yes	Yes	Yes
Observations	1,711	1,711	1,792	1,792
R-squared	0.095	0.107	0.150	0.144
Panel B: Conceptual knowledge treatment (T_i)				
	Chinese market		US market	
	(1)	(2)	(3)	(4)
$T_i \times Accruals$	-0.041*** (-2.66)		-0.037 (-0.71)	
$Accruals$	0.020*** (3.20)		-0.059 (-1.54)	
$T_i \times Accruals_{UD}$		-0.041** (-2.27)		-0.062* (-1.84)
$Accruals_{UD}$		0.058*** (5.70)		-0.012 (-0.45)
T_i	-0.008*** (-3.01)	-0.005* (-1.81)	-0.004 (-1.22)	-0.002 (-0.65)
SUE	0.647*** (5.51)	0.607*** (5.21)	0.016 (1.31)	0.012 (0.98)
Industry and month FEs	Yes	Yes	Yes	Yes
Observations	1,142	1,142	1,193	1,193
R-squared	0.112	0.129	0.183	0.184

Panel C: Conceptual and methodological knowledge treatment (T_2)				
	Chinese market		US market	
	(1)	(2)	(3)	(4)
$T_2 \times Accruals$	-0.030*** (-3.34)		-0.155*** (-2.85)	
$Accruals$	0.020*** (3.32)		-0.080** (-2.07)	
$T_2 \times Accruals_{UD}$		-0.072*** (-3.87)		-0.072** (-2.17)
$Accruals_{UD}$		0.059*** (6.22)		-0.007 (-0.26)
T_2	-0.016*** (-6.44)	-0.013*** (-5.26)	-0.010*** (-2.68)	-0.000 (-0.12)
SUE	0.459*** (4.00)	0.435*** (3.83)	0.021 (0.86)	0.035 (1.43)
Industry and month FEs	Yes	Yes	Yes	Yes
Observations	1,142	1,142	1,193	1,193
R-squared	0.121	0.141	0.212	0.197

Notes: The table reports the effects of education treatment on accrual pricing in the short-term window, using group C as the control. The sample in Panel A consists of stocks in the treatment groups T_1 and T_2 and the control group C. The sample in Panel B (C) consists of stocks in the treatment groups T_1 (T_2) and the control group C. The dependent variable is $CAR(0,1)$, the cumulative abnormal return from day 0 to day 1 (day 0 is EA day), is based on the market model. $Treat$ equals one if a stock is in T_1 or T_2 and zero if it is in C. T_1 equals one if a stock is in T_1 and zero if it is in C. T_2 equals one if a stock is in T_2 and zero if it is in C. $Accruals$ is the change in current assets minus the change in cash holding minus the change in current liabilities plus the change in short-term debt plus depreciation, scaled by total assets. $Accruals_{UD}$ is discretionary accruals in the current year minus discretionary accruals in the previous year. Discretionary accruals are residuals from regressing firms' total accruals (i.e., $Accruals$) on the reciprocal of total assets; the change in sales/total assets; net property, plant, and equipment/total assets; and return on assets in each industry-year. SUE measures EA news, which is the announced earnings per share minus the consensus of analyst forecasts in the past year, scaled by the stock price before the EAs. The fixed effects of industry and EA months are included. The t -statistics are shown in brackets. The significance levels at 1%, 5%, and 10% is indicated by ***, **, and *, respectively.

Table 5. Portfolio analysis
Panel A: Chinese market

A1: Size/BM adjusted returns				
Portfolio	(1)	(2)	(3)	(4)
Accruals ranking	S	C	T ₁	T ₂
Low	0.055*	0.095**	0.080**	-0.028
High	-0.017	-0.022	0.010	0.005
High – Low	-0.072*	-0.117**	-0.070	0.033
A2: Alpha based on Fama–French three-factor model				
Portfolio	(1)	(2)	(3)	(4)
Accruals ranking	S	C	T ₁	T ₂
Low	0.097***	0.119***	0.100**	0.020
High	0.003	0.052	0.015	0.005
High – Low	-0.126**	-0.168**	-0.030	0.024

Panel B: US market

B1: Size/BM adjusted returns				
Portfolio	(1)	(2)	(3)	(4)
Accruals ranking	S	C	T ₁	T ₂
Low	0.044**	0.198***	-0.023	-0.001
High	-0.031***	0.002	0.018	0.051***
High – Low	-0.075***	-0.196***	0.041*	0.052**
B2: Alpha based on Fama–French three-factor model				
Portfolio	(1)	(2)	(3)	(4)
Accruals ranking	S	C	T ₁	T ₂
Low	0.052***	0.118***	0.048*	0.016
High	-0.002	0.040*	0.056**	0.065**
High – Low	-0.054**	-0.079**	0.008	0.048

Notes: This table presents the abnormal returns of portfolios formed based on unexpected discretionary accruals ($Accruals_{UD}$). The sample consists of stocks in the treatment groups T₁ and T₂, the control group C, and the comparison group S. $Accruals_{UD}$ is discretionary accruals in the current year minus discretionary accruals in the previous year. Discretionary accruals are residuals from regressing firms' total accruals (i.e., $Accruals$) on the reciprocal of total assets; the change in sales/total assets; net property, plant, and equipment/total assets; and return on assets in each industry-year, where $Accruals$ is the change in current assets minus the change in cash holdings minus the change in current liabilities plus the change in short-term debt plus depreciation, scaled by total assets. The “Low” and “High” accrual portfolios consist of stocks with $Accruals_{UD}$ in the bottom and top deciles of each group, respectively. To calculate the Size/BM adjusted returns, we first calculate stocks' abnormal returns (raw returns minus the returns of the matched portfolio of 5×5 size, and book-to-market portfolios). We then calculate a portfolio's daily abnormal returns by computing the value-weighted average of abnormal returns of stocks in the portfolio. Finally, the Size/BM adjusted return is the average of a portfolio's abnormal returns from day 11 to day 251 (day 0 is the EA date for fiscal year 2019). To calculate alpha based on the Fama–French three-factor model, we first calculate a portfolio's daily returns by computing the value-weighted average of raw returns of the stocks in the portfolio. The alpha based on the Fama–French three-factor model is the constant obtained from the regression of a portfolio's daily excess returns (return – risk-free rate) on the Fama–French three factors from day 11 to day 251. All of the returns are in daily percentages. Significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *, respectively.

Table 6. Financial education and accrual pricing in a long-term window

Panel A: Average treatment ($Treat$)				
	Chinese market		US market	
	(1)	(2)	(3)	(4)
$Treat \times Accruals$	0.278*** (3.21)		0.775** (2.24)	
$Accruals$	-0.347*** (-5.74)		-0.871*** (-3.08)	
$Treat \times Accruals_{UD}$		0.844*** (5.77)		0.412* (1.87)
$Accruals_{UD}$		-0.628*** (-6.41)		-0.223 (-1.16)
$Treat$	-0.073*** (-3.53)	-0.094*** (-4.49)	-0.043* (-1.80)	-0.095*** (-6.28)
SUE	1.880** (2.01)	2.095** (2.25)	-0.023 (-0.26)	-0.005 (-0.06)
Industry and month FEs	Yes	Yes	Yes	Yes
Observations	1,711	1,711	1,784	1,784
R-squared	0.136	0.142	0.244	0.242
Panel B: Conceptual knowledge treatment (T_i)				
	Chinese market		US market	
	(1)	(2)	(3)	(4)
$T_i \times Accruals$	-0.139 (-0.94)		-0.023 (-0.06)	
$Accruals$	-0.372*** (-6.10)		-0.517* (-1.72)	
$T_i \times Accruals_{UD}$		0.734*** (4.09)		0.205 (0.77)
$Accruals_{UD}$		-0.655*** (-6.60)		-0.154 (-0.76)
T_i	-0.021 (-0.83)	-0.037 (-1.47)	-0.040 (-1.39)	-0.045** (-2.40)
SUE	1.973* (1.73)	2.088* (1.83)	-0.036 (-0.38)	-0.020 (-0.21)
Industry and month FEs	Yes	Yes	Yes	Yes
Observations	1,142	1,142	1,186	1,186
R-squared	0.179	0.173	0.253	0.250

Panel C: Conceptual and methodological knowledge treatment (T_2)				
	Chinese market		US market	
	(1)	(2)	(3)	(4)
$T_2 \times Accruals$	0.380*** (4.14)		1.282*** (3.21)	
$Accruals$	-0.319*** (-5.30)		-0.862*** (-3.04)	
$T_2 \times Accruals_{UD}$		1.124*** (5.97)		0.533** (2.19)
$Accruals_{UD}$		-0.601*** (-6.21)		-0.065 (-0.34)
T_2	-0.117*** (-4.73)	-0.143*** (-5.79)	-0.035 (-1.29)	-0.115*** (-6.69)
SUE	2.129* (1.83)	2.287** (1.99)	0.181 (1.02)	0.168 (0.94)
Industry and month FEs	Yes	Yes	Yes	Yes
Observations	1,142	1,142	1,190	1,190
R-squared	0.190	0.204	0.336	0.334

Notes: This table reports the effects of the education treatment on accrual pricing in the long-term window, using group C as the control. The sample in Panel A consists of stocks in the treatment groups T_1 and T_2 and the control group C. The sample in Panel B (C) consists of stocks in the treatment groups T_1 (T_2) and the control group C. The dependent variable is $CAR(11,251)$, the buy-and-hold return from day 11 to day 251 (day 0 is the EA date) minus the return of the matched portfolio of 5×5 size and book-to-market portfolios in the same window. $Treat$ equals one if a stock is in T_1 or T_2 and zero if it is in C. T_1 equals one if a stock is in T_1 and zero if it is in C. T_2 equals one if a stock is in T_2 and zero if it is in C. $Accruals$ is the change in current assets minus the change in cash holding minus the change in current liabilities plus the change in short-term debt plus depreciation, scaled by total assets. $Accruals_{UD}$ is discretionary accruals in the current year minus discretionary accruals in the previous year. Discretionary accruals are residuals from regressing firms' total accruals (i.e., $Accruals$) on the reciprocal of total assets; the change in sales/total assets; net property, plant, and equipment/total assets; and return on assets in each industry-year. SUE measures EA news, which is the announced earnings per share minus the consensus of analyst forecasts in the past year, scaled by the stock price before the EAs. The fixed effects of industries and EA months are included. The t -statistics are shown in brackets. Significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *, respectively.

Table 7. Financial education effects and reading volume

Panel A: Average treatment ($Treat$)						
	Chinese market			US market		
	(1)	(2)	(3)	(4)	(5)	(6)
$Treat \times Accruals_{UD} \times Reads$	-0.229** (-2.43)			-0.056 (-0.92)		
$Treat \times Reads$	-0.002 (-0.20)			0.001 (0.24)		
$Accruals_{UD} \times Reads$	0.080 (0.87)			-0.032 (-0.62)		
$Reads$	0.012 (1.01)			0.011*** (2.92)		
$Treat \times Accruals_{UD} \times CSLS$		-0.201*** (-5.59)			-0.008 (-0.12)	
$Treat \times CSLS$		0.002 (0.31)			-0.007 (-1.40)	
$Accruals_{UD} \times CSLS$		0.056* (1.96)			-0.060 (-0.99)	
$CSLS$		-0.001 (-0.22)			0.009* (1.88)	
$Treat \times Accruals_{UD} \times WebView$			-0.124*** (-4.59)			-0.069** (-2.29)
$WebView$			0.002 (0.46)			-0.002 (-0.74)
$Treat \times Accruals_{UD}$	0.121 (1.30)	0.068** (2.20)	-0.033** (-2.15)	-0.012 (-0.26)	-0.056 (-0.95)	-0.017 (-0.50)
$Accruals_{UD}$	-0.021 (-0.23)	0.012 (0.44)	0.058*** (6.03)	0.010 (0.24)	0.039 (0.72)	-0.008 (-0.32)
$Treat$	-0.003 (-0.27)	-0.009** (-2.09)	-0.009*** (-4.09)	-0.001 (-0.29)	0.005 (1.09)	0.000 (0.10)
SUE	0.456*** (5.00)	0.446*** (4.86)	0.455*** (4.92)	0.021* (1.67)	0.024** (1.96)	0.017 (1.34)
Industry and month FEs	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,711	1,711	1,711	1,792	1,792	1,792
R-squared	0.138	0.133	0.119	0.160	0.148	0.147
Panel B: Conceptual knowledge treatment (T_i)						
	Chinese market			US market		
	(1)	(2)	(3)	(4)	(5)	(6)
$T_i \times Accruals_{UD} \times Reads$	-0.281*** (-2.82)			0.007 (0.10)		
$T_i \times Reads$	-0.007 (-0.54)			-0.007 (-1.35)		
$Accruals_{UD} \times Reads$	0.081 (0.86)			-0.012 (-0.25)		
$Reads$	0.012 (0.95)			0.011*** (2.99)		
$T_i \times Accruals_{UD} \times CSLS$		-0.220*** (-5.17)			-0.137* (-1.84)	
$T_i \times CSLS$		-0.003 (-0.47)			-0.001 (-0.14)	

$Accruals_{UD} \times CSLS$			0.053*			-0.070
			(1.77)			(-1.18)
$CSLS$			-0.001			0.006
			(-0.22)			(1.40)
$T_1 \times Accruals_{UD}$ $\times WebView$				-0.094***		-0.116**
				(-2.64)		(-2.42)
$WebView$				0.004		-0.002
				(0.92)		(-0.49)
$Treat \times Accruals_{UD}$	0.178*	0.088**	-0.018	-0.063	0.022	0.003
	(1.82)	(2.50)	(-0.92)	(-1.10)	(0.35)	(0.06)
$Accruals_{UD}$	-0.024	0.014	0.057***	-0.006	0.044	-0.011
	(-0.26)	(0.51)	(5.68)	(-0.14)	(0.84)	(-0.44)
T_1	0.004	-0.003	-0.007**	0.004	-0.001	-0.001
	(0.30)	(-0.49)	(-2.12)	(0.81)	(-0.11)	(-0.28)
SUE	0.633***	0.607***	0.611***	0.013	0.022*	0.003
	(5.52)	(5.27)	(5.26)	(1.05)	(1.83)	(0.28)
Industry and month FEs	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,142	1,142	1,142	1,193	1,193	1,193
R-squared	0.161	0.155	0.135	0.192	0.198	0.190

Panel C: Conceptual and methodological knowledge treatment (T_2)

	Chinese market			US market		
	(1)	(2)	(3)	(4)	(5)	(6)
$T_2 \times Accruals_{UD} \times Reads$	-0.182*			-0.131*		
	(-1.90)			(-1.96)		
$T_2 \times Reads$	-0.002			0.002		
	(-0.18)			(0.32)		
$Accruals_{UD} \times Reads$	0.076			-0.038		
	(0.84)			(-0.73)		
$Reads$	0.013			0.013***		
	(1.08)			(3.39)		
$T_2 \times Accruals_{UD} \times CSLS$		-0.177***			0.107	
		(-4.16)			(1.42)	
$T_2 \times CSLS$		0.004			-0.015***	
		(0.72)			(-2.63)	
$Accruals_{UD} \times CSLS$		0.059**			-0.069	
		(2.08)			(-1.13)	
$CSLS$		-0.000			0.010**	
		(-0.07)			(2.06)	
$T_2 \times Accruals_{UD}$ $\times WebView$			-0.066*			-0.041
			(-1.76)			(-0.95)
$WebView$			0.001			-0.002
			(0.32)			(-0.59)
$T_2 \times Accruals_{UD}$	0.070	0.046	-0.056***	0.012	-0.150**	-0.054
	(0.75)	(1.30)	(-2.71)	(0.22)	(-2.35)	(-1.41)
$Accruals_{UD}$	-0.017	0.010	0.059***	0.011	0.049	-0.007
	(-0.18)	(0.36)	(6.20)	(0.27)	(0.90)	(-0.28)
T_2	-0.007	-0.014***	-0.013***	0.000	0.012**	0.001
	(-0.57)	(-3.09)	(-4.82)	(0.08)	(2.28)	(0.35)
SUE	0.421***	0.423***	0.434***	0.032	0.033	0.037
	(3.72)	(3.73)	(3.82)	(1.34)	(1.34)	(1.51)

Industry and month FEs	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,142	1,142	1,142	1,193	1,193	1,193
R-squared	0.156	0.156	0.144	0.221	0.202	0.198

Notes: The sample in Panel A consists of stocks in the treatment groups T_1 and T_2 and the control group C. The sample in Panel B (C) consists of stocks in the treatment groups T_1 (T_2) and the control group C. The dependent variable is $CAR(0,1)$, the cumulative abnormal return from day 0 to day 1 (day 0 is EA day) based on the market model. $Treat$ equals one if a stock is in T_1 or T_2 and zero if it is in C. T_1 equals one if a stock is in T_1 and zero if it is in C. T_2 equals one if a stock is in T_2 and zero if it is in C. $Accruals_{UD}$ is discretionary accruals in the current year minus discretionary accruals in the previous year. Discretionary accruals are residuals from regressing firms' total accruals (i.e., $Accruals$) on the reciprocal of total assets; the change in sales/total assets; net property, plant, and equipment/total assets; and return on assets in each industry-year. $Reads$ is set to one if the total number of posts read across all social media platforms for a stock exceeds the bottom tercile, and zero otherwise. $CSLS$ is set to one if the combined number of comments or replies on our posts for a specific stock across all platforms exceeds the bottom tercile, or if the posts are "retweeted/shared," "liked," or "saved" on any platforms, and zero otherwise. $WebView$ is set to one if the number of clicks on both type 1 and type 2 link pages for a stock surpasses the bottom tercile among the treatment stocks, and zero otherwise. SUE is earnings per share minus the consensus of analyst forecasts in the past year, scaled by the stock price before the EAs. The interaction terms $Treat \times Webview$ and $Accruals_{UD} \times Webview$ are omitted to avoid multicollinearity because $WebView$ indicates a subgroup of the treatment stocks and takes the value of zero in all of the control stocks (the stocks in group C do not have link pages). The t -statistics are shown in brackets. Significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *, respectively.

Table 8. Financial education effects and retail investors

Panel A: Average treatment ($Treat$)		
	Chinese market	US market
	(1)	(2)
$Treat \times Accruals_{UD} \times RI$	-0.046*** (-3.04)	-0.095 (-1.34)
$Treat \times RI$	0.002 (0.35)	0.007 (1.23)
$Accruals_{UD} \times RI$	0.005 (0.44)	0.117** (1.97)
RI	-0.008* (-1.76)	0.001 (0.13)
$Treat \times Accruals_{UD}$	-0.029** (-2.40)	0.017 (0.28)
$Accruals_{UD}$	0.056*** (3.31)	-0.094* (-1.84)
$Treat$	-0.008*** (-3.27)	-0.006 (-1.20)
SUE	0.445*** (3.47)	0.021* (1.73)
Industry and month FEs	Yes	Yes
Observations	1,711	1,792
R-squared	0.114	0.150
Panel B: Conceptual knowledge treatment (T_i)		
	Chinese market	US market
	(1)	(2)
$T_i \times Accruals_{UD} \times RI$	-0.128*** (-2.95)	-0.068 (-0.86)
$T_i \times RI$	-0.010 (-1.09)	0.002 (0.37)
$Accruals_{UD} \times RI$	0.002 (0.18)	0.080 (1.39)
RI	-0.006 (-1.39)	-0.004 (-0.95)
$T_i \times Accruals_{UD}$	0.044 (1.33)	-0.012 (-0.18)
$Accruals_{UD}$	0.055*** (3.38)	-0.071 (-1.42)
T_i	0.005 (0.75)	-0.003 (-0.58)
SUE	0.606*** (7.86)	0.012 (0.97)
Industry and month FEs	Yes	Yes
Observations	1,142	1,193
R-squared	0.152	0.186

Panel C: Conceptual and methodological knowledge treatment (T_2)

	Chinese market	US market
	(1)	(2)
$T_2 \times Accruals_{UD} \times RI$	0.002 (0.04)	-0.096 (-1.23)
$T_2 \times RI$	0.009** (2.41)	0.009 (1.49)
$Accruals_{UD} \times RI$	0.008 (0.90)	0.118** (2.01)
RI	-0.008* (-1.78)	0.004 (0.78)
$T_2 \times Accruals_{UD}$	-0.074* (-1.93)	0.002 (0.02)
$Accruals_{UD}$	0.056*** (3.28)	-0.095* (-1.88)
T_2	-0.016*** (-10.32)	-0.006 (-1.16)
SUE	0.428*** (4.54)	0.035 (1.42)
Industry and month FEs	Yes	Yes
Observations	1,142	1,193
R-squared	0.145	0.210

Notes: The sample consists of stocks in the treatment groups T_1 and T_2 and control group C. The sample in Panel B (C) consists of stocks in the treatment groups T_1 (T_2) and the control group C. The dependent variable is $CAR(0,1)$, the cumulative abnormal return from day 0 to day 1 (day 0 is the EA day) based on the market model. $Treat$ equals one if a stock is in T_1 or T_2 and zero if it is in C. T_1 equals one if a stock is in T_1 and zero if it is in C. T_2 equals one if a stock is in T_2 and zero if it is in C. $Accruals_{UD}$ is discretionary accruals in the current year minus discretionary accruals in the previous year. Discretionary accruals are residuals from regressing firms' total accruals (i.e., $Accruals$) on the reciprocal of total assets; the change in sales/total assets; net property, plant, and equipment/total assets; and return on assets in each industry-year. RI is assigned a value of one if the intensity of retail investors exceeds the bottom tercile of our sample, and zero otherwise. We measure the intensity of retail investors using 100% minus the number of shares held by institutional investors over the total number of tradable shares. SUE measures the EA surprise, which is the announced earnings per share minus the consensus of analyst forecasts in the past year, scaled by the stock price before the EAs. The fixed effects of industries and EA months are included. The t -statistics are shown in brackets. Significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *, respectively.

Internet Appendix
to
**“Knowledge is Power:
A Field Experiment in the Chinese and US Stock Markets”**

Internet Appendix 1.1. Conceptual knowledge of accruals

Internet Appendix 1.2. Methodological knowledge of accruals

Internet Appendix 1.3. Spreadsheet module for estimating accruals and abnormal accruals

Internet Appendix 2. Sample of initial posts on social media

Internet Appendix 3. Posting schedule on social media

Internet Appendix 4. Daily experiment activities during the experiment window among treatment groups T₁ and T₂ and control group C

Internet Appendix 5. Samples of post responses

Internet Appendix 6. Testing for accrual mispricing

Internet Appendix 7. Financial education and accrual pricing: Placebo test

Internet Appendix 8. Financial education and accrual pricing in the short-term window: Group S as the control

Internet Appendix 9. Financial education and accrual pricing in a long-term window: Group S as the control

Internet Appendix 10. Financial education effects and reading volume: Group S as the control

Internet Appendix 11. Financial education effects and retail investors: Group S as the control

Internet Appendix 12. Compare control and comparison groups

Internet Appendix 13. Compare the pricing of positive and negative accruals

Internet Appendix 1.1. Conceptual knowledge of accruals

Panel A: English version

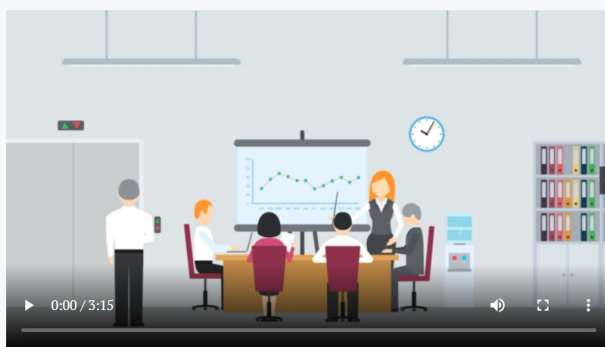
Excerpt from “Conceptual Knowledge of Accruals” for Coca Cola (Ticker: COKE). The full article is available at: <http://www.financial-education-hub.com/1/COKE>.

Highlight

Firms' reported earnings often differ from their actual cash flows. This difference is referred to as "accruals" and represents the component of earnings that have not yet been received. Due to estimation errors, the value of accruals is not necessarily equal to the amount of cash that is received, causing high accruals in the current period in relation to low earnings in subsequent periods. This effect is called low earnings persistence of accruals.

Example

In 2019, company A reports earnings of \$100 million (i.e., operating income after depreciation). Cash flows and accruals are \$20 million and \$80 million, respectively. Company B also reports earnings of \$100 million. Its cash flows and accruals are \$70 million and \$30 million, respectively. Based on the concept of low earnings persistence of accruals, holding all other factors constant, A is expected to have lower earnings than B in 2020.



Panel B: Chinese version

Excerpt from “Conceptual Knowledge of Accruals” for Shanghai New World Co. (Ticker: 600628). The full article is available at: <http://www.financial-education-hub.com/1/600628>.

摘要

公司报告的利润经常与实际的现金流不同。这个差异被称为“应计项目”，代表了尚未收到现金的利润部分。由于估计误差，应计项目的金额可能不等于未来收到现金的金额，导致当期应计项目高的公司后期的收益会更低。这种效应被称为应计项对盈利的低持续性。

例子

2019年，A公司公布的利润是1亿元（折旧费用后的营业收入）。现金流和应计项目部分分别占2 000万元和8 000万元。公司B公布的利润也是1亿元。现金流和应计项目部分分别占7 000万元和3 000万元。

根据应计项目低持续性的特性，在其他方面都相同的情况下，可以预计A公司在2020年的收益将会低于B公司。



Internet Appendix 1.2. Methodological knowledge of accruals

Panel A: English version

Excerpt from “Methodological Knowledge of Accruals” for Coca Cola (Ticker: COKE). The full article is available at: <http://www.financial-education-hub.com/2/COKE>.

Estimate Accruals and Earnings Quality

High accruals may associate with lower subsequent earnings (i.e., low earnings persistence of accruals; please see [Coca Cola Consolidated Inc](#)). This page gives you a quick way to compute the accruals of **Coca Cola Consolidated Inc** and identify the quality of the firm's reported earnings, which will help you to evaluate the firm's future performance.

A spreadsheet module

A spreadsheet module is provided for **Coca Cola Consolidated Inc** to calculate its accruals, error terms, and the standard deviation of the errors. Please click [Accrual Calculator](#) to download the module. The definition of accruals is provided in **Appendix A** of this web page. A larger error term and higher standard deviation of the error terms means poorer quality of accruals and reported earnings (see **Appendix B** for detailed introduction).

To view the results, please input the 2019 financial information based on the firm's just announced financial report. That is, update the information in the **green cells** as shown below. Information for other years is already included.

	[1]	[2]	[3]	[4]	[5]	[6]	[7]
	million \$						
Year (t)	Current assets (CA)	Current liabilities (CL)	Cash and short-term investments (Cash)	Debts in current liabilities (Debt)	Total assets (TA)	Cash flow from operating activities (CF)	Earnings (operating income after depreciation) (E)
	Source: balance sheet	Source: balance sheet	Source: balance sheet	Source: balance sheet	Source: balance sheet	Source: Statement of Cash Flow	Source: Income Statement
2019							

After inputting the information, you will see the results on the bottom right-hand side of the sheet (see example below), including the firm's 2019 accruals, average accruals over the last ten years, 2019 accruals error, and the standard deviation of accruals errors over the last ten years. Moreover, the position of these statistics in the firm's industry and market are also presented. **Q1** indicates a very low level (1st quartile, or the bottom 25%), **Q2** indicates a low level (2nd quartile), **Q3** indicates a high level (3rd quartile), and **Q4** indicates a very high level (4th quartile or top 25%).

Panel B: Chinese version

Excerpt from “Methodological Knowledge of Accruals” for Shanghai New World Co. (Ticker: 600628). The full article is available at: <http://www.financial-education-hub.com/2/600628>.

估计“应计项目”和利润的质量

当期“应计项目”较高的公司其后期往往会录得较低的收益（这个特性被称为“应计项目”的低持续性，请参看 [新世界](#)）。这个页面提供一个快速计算 **新世界** “应计项目”的方法，从而确定其报告收益的质量，并帮助你评估此公司未来的业绩。

电子表格模板

这里提供了一个电子表格模板，专门为 **新世界** 用于计算其应计项目具体水平、应计项目与实际现金流之间的误差项，以及这些误差项的方法。请点击 [应计项目计算器](#) 下载这个模板。应计项目的定义见本网页 **附录 1**。当误差项越大或误差项的标准差越大时，意味着其应计项目和报告收益的质量就越差（**附录 2**见详细的介绍）。

查看结果前，请先根据公司刚刚公布的财务报告输入2019年的财务信息。也就是请更新如下所示绿色单元格中的信息。其他年份的资料已经包括在表格内。

	[1]	[2]	[3]	[4]	[5]	[6]	[7]
	单位: 百万元人民币						
年 (t)	流动资产 (CA)	流动负债 (CL)	现金及短期投资 (Cash)	流动负债中的债务部分 (Debt)	总资产 (TA)	经营活动现金流量 (CF)	利润 (折旧后营业收入) (E)
	来源: 资产负债表	来源: 资产负债表	来源: 资产负债表	来源: 资产负债表	来源: 资产负债表	来源: 现金流量表	来源: 损益表
2019							

输入信息后，你将在右下角看到结果(如下图所示)。其包括公司2019年应计项目的水平，过去十年应计项目的平均水平，2019年应计项目与实际现金流之间的误差项，以及过去十年这个误差项的方差。此外，还给出了公司这些指标在其所在行业和整个市场中的位置。**Q1**表示本指标处于非常低的水平(在最底部的25%)，**Q2**表示较低水平(处于25%-50%之间)，**Q3**表示较高水平(处于50%-75%之间)，**Q4**表示非常高水平(处于最顶部的75%)。

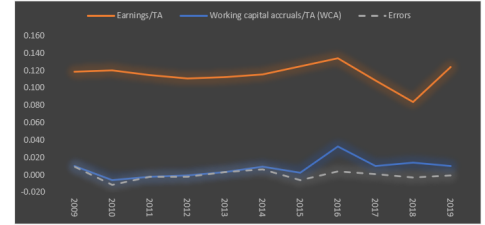
Internet Appendix 1.3. Spreadsheet module for estimating accruals and abnormal accruals

Panel A: English version

Module for Coca Cola (Ticker: COKE)

Notes: Please update the firm's financial information in 2019 in the green cells, the analysis results will be updated accordingly

Firm Ticker:	COKE	Firm:	Coca Cola Consolidated Inc		Industry:	Candy & Soda		4
		[1]	[2]	[3]	[4]	[5]	[6]	[7]
		million \$						
obs.	Year (t)	Current assets (CA)	Current liabilities (CL)	Cash and short-term investments (Cash)	Debts in current liabilities (Debt)	Total assets (TA)	Cash flow from operating activities (CF)	Earnings (operating income before depreciation) (E)
		Source: balance sheet	Source: balance sheet	Source: balance sheet	Source: balance sheet	Source: balance sheet	Source: Statement of Cash Flow	Source: Income Statement
	2019	830	622	10	24	3,127	290	374
1	2019	830.3	622.2	9.6	24.4	3126.9	290.4	374.3
2	2018	797.4	601.7	13.5	8.6	3009.9	168.9	256.8
3	2017	794.5	639.5	16.9	8.2	3073.0	307.8	266.7
4	2016	593.8	457.9	21.9	7.5	2449.5	162.0	248.5
5	2015	436.0	326.6	55.5	7.1	1850.8	108.3	179.0
6	2014	287.0	227.4	9.1	6.4	1433.1	91.9	147.1
7	2013	239.2	208.8	11.8	25.9	1276.2	96.4	144.3
8	2012	241.3	216.3	10.4	25.2	1283.5	83.2	150.5
9	2011	311.8	302.5	93.8	124.6	1361.2	109.7	149.6
10	2010	264.7	176.7	49.4	3.9	1307.6	98.1	154.6
11	2009	230.2	162.0	22.3	3.8	1283.1	79.5	155.9
12	2008	248.3	346.1	45.4	179.5	1315.8	96.6	146.4
13	2007	198.3	159.3	9.9	10.0	1291.8	95.5	152.5



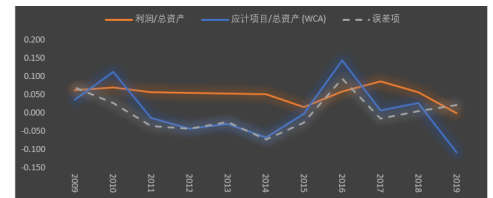
The firm's accrual level and quality			The firm's position in the industry or market: Q1-very low (1st quarter, bottom 25%), Q2-low (2nd quarter), Q3-high (3rd quarter), and Q4-very high (4th quarter)	
			industry	market
Accrual level	Accrual in 2019	0.011	Q2	Q3
	Average accruals over the past 10 years	0.008	Q3	Q3
Accrual quality	Accrual error in 2019	0.000	Q3	Q2
	Standard deviation of errors over the past 10 years	0.006	Q1	Q1

Panel B: Chinese version

Module for Shanghai New World Co. (Ticker: 600628)

注意: 请在绿色部分输入本公司2019年的财务信息, 分析结果将自动更新

股票代码:	600628	公司:	新世界	行业:	零售业	10		
		[1]	[2]	[3]	[4]	[5]	[6]	[7]
		单位: 百万元人民币						
观测值	年 (t)	流动资产 (CA)	流动负债 (CL)	现金及短期投资 (Cash)	流动负债中的债务部分 (Debt)	总资产 (TA)	经营活动现金流量 (CF)	利润 (营业收入) (E)
		来源: 资产负债表	来源: 资产负债表	来源: 资产负债表	来源: 资产负债表	来源: 资产负债表	来源: 现金流量表	来源: 损益表
	2019	2,533	2,132	1,575	800	6,571	63	1.04
1	2019	2532.8	2132.3	1575.0	800.0	6570.5	62.8	1.0
2	2018	2548.0	1132.2	1364.3	200.0	5793.3	330.2	332.2
3	2017	2469.8	1384.3	1273.4	276.0	5856.0	417.9	498.3
4	2016	2670.8	1707.1	1663.1	747.0	5782.3	457.6	282.0
5	2015	1085.9	2142.4	677.7	1089.0	4784.8	500.0	82.7
6	2014	1237.6	2565.2	572.1	1268.0	5214.6	466.9	270.7
7	2013	1383.1	2765.4	670.4	1770.0	5226.7	414.8	277.1
8	2012	1245.1	2919.9	533.5	2081.0	5198.3	445.5	312.6
9	2011	1545.7	3460.9	737.1	2759.0	5568.3	383.2	284.9
10	2010	1258.9	3086.7	388.6	2393.0	5043.5	477.5	273.3
11	2009	554.2	2123.4	220.1	1523.5	3943.5	308.3	251.8
12	2008	502.1	2283.3	289.2	1662.0	3961.9	371.2	229.9
13	2007	467.7	2380.1	251.4	1754.3	4007.5	442.6	212.0



公司应计项目的水平和质量			公司指标在行业或市场中的位置: Q1-非常低(低于25%), Q2-比较低(25%-50%), Q3-比较高(50%-75%), Q4-非常高(高于75%)	
			行业	市场
应计项目 的水平	2019年应计项目的水平	-0.108	Q1	Q1
	过去10年应计项目的平均水平	0.006	Q2	Q2
应计项目 的质量	2019年应计项目估计误差	0.021	Q3	Q3
	过去十年应计项目估计误差的 方差	0.050	Q3	Q2

Internet Appendix 2. Sample of initial posts on social media

Panel A: English version

Group	Sample
C	\$Ticker will announce its 2019 annual performance on March 15, 2020.
T ₁	\$Ticker will announce its 2019 annual performance on March 15, 2020. Please note that a company may report high earnings but actually have low cash flows. This difference is called “accruals,” which often refers to revenues or expenses that do not involve cash flows, but are included in the firm’s current earnings. Accruals are estimated by the company, which may involve estimation errors. Some companies may even use accruals to manipulate earnings. As a result, high accruals may not result in high subsequent cash flows. Investors focusing on bottom-line earnings as reported by the company, but ignoring the quality of the earnings, may misunderstand the company’s true performance. For example, if two companies report the same level of earnings, the one with high accruals would be more likely to feature lower profitability in the following year. For more knowledge on the nature and effects of accruals, please see: http://www.financial-education-hub.com/1/\$Ticker .
T ₂	\$Ticker will announce its 2019 annual performance on March 15, 2020. Please note that a company may report high earnings but actually have low cash flows. This difference is called “accruals,” which often refers to revenues or expenses that do not involve cash flows, but are included in the current earnings. Accruals are estimated by the company, which may involve estimation errors. Some companies may even use accruals to manipulate earnings. As a result, high accruals may not result in high subsequent cash flows. Investors focusing on bottom-line earnings as reported by the company, but ignoring the quality of the earnings, may thus misunderstand the company’s true performance. For example, if two companies report the same level of earnings, the one with high accruals is more likely to report lower profitability in the following year. For more knowledge on the nature and effects of accruals, please see: http://www.financial-education-hub.com/1/\$Ticker . If you want to estimate the quality of \$Ticker’s earnings that are released, you can calculate the accruals level, and the portion that may not be realized in cash flows. For details on this method, please see: http://www.financial-education-hub.com/2/\$Ticker .

Panel B: Chinese version

Groups	Sample
C	\$Ticker 将于 2020 年 3 月 15 日公布 2019 全年业绩。
T ₁	\$Ticker 将于 2020 年 3 月 15 日公布 2019 全年业绩。请注意，有些公司公布的账面利润可能很高但实际的现金流却很少。这个差额叫做“应计项目”，通常是指那些不直接形成当前现金流却计入当前损益的收入或费用。“应计项目”需要通过公司的估计得出，通常会出现估算错误。一些公司甚至利用“应计项目”来操纵盈利。因此，公告利润的“应计项目”部分在未来未必能兑现成实际的现金流。为此，投资者只关注公司所公布的利润而忽视其质量，可能会误读公司业绩。比如，两家当前利润一样的公司，“应计项目”高的那家公司下一年的盈利可能会更低。关于这方面的具体的知识： http://www.financial-education-hub.com/1/\$Ticker .
T ₂	\$Ticker 将于 2020 年 3 月 15 日公布 2019 全年业绩。请注意，有些公司公布的账面利润可能很高但实际的现金流却很少。这个差额叫做“应计项目”，通常是指那些不直接形成当前现金流却计入当前损益的收入或费用。“应计项目”需要通过公司的估计得出，通常会出现估算错误。一些公司甚至利用“应计项目”来操纵盈利。因此，公告利润的“应计项目”部分在未来未必能兑现成实际的现金流。为此，投资者只关注公司所公布的利润而忽视其质量，可能会误读公司业绩。比如，两家当前利润一样的公司，“应计项目”高的那家公司下一年的盈利可能会更低。关于这方面的具体的知识： http://www.financial-education-hub.com/1/\$Ticker 。如想估算\$Ticker 所公布利润的质量，可以计算其“应计项目”的具体水平，以及不能兑现成现金流的部分，具体方法： http://www.financial-education-hub.com/2/\$Ticker .

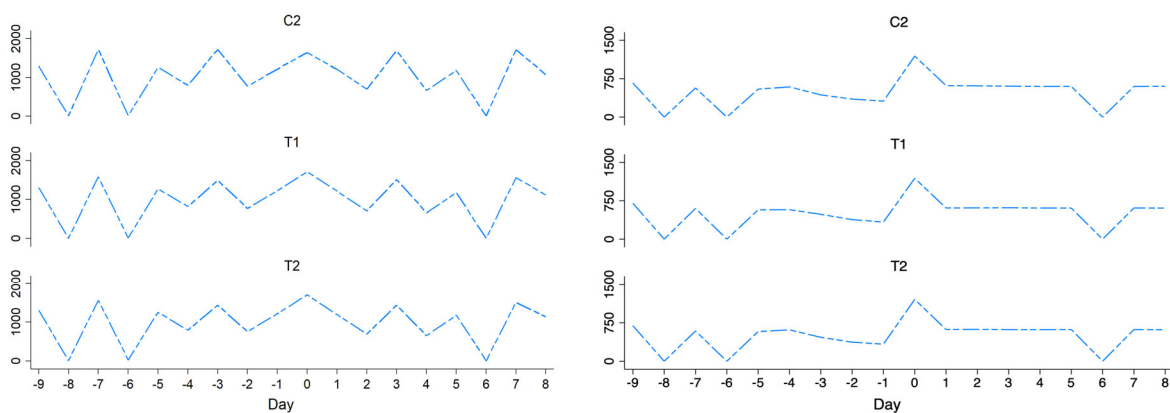
Internet Appendix 3. Posting schedule on social media

Day	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	1	2	3	4	5	6	7	8	9
Is there a post? (Y if yes)	Y		Y		Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y		Y		Y
Post No.	1		2		3	4	5	6	7	8	9	10	11	12	13		14		15

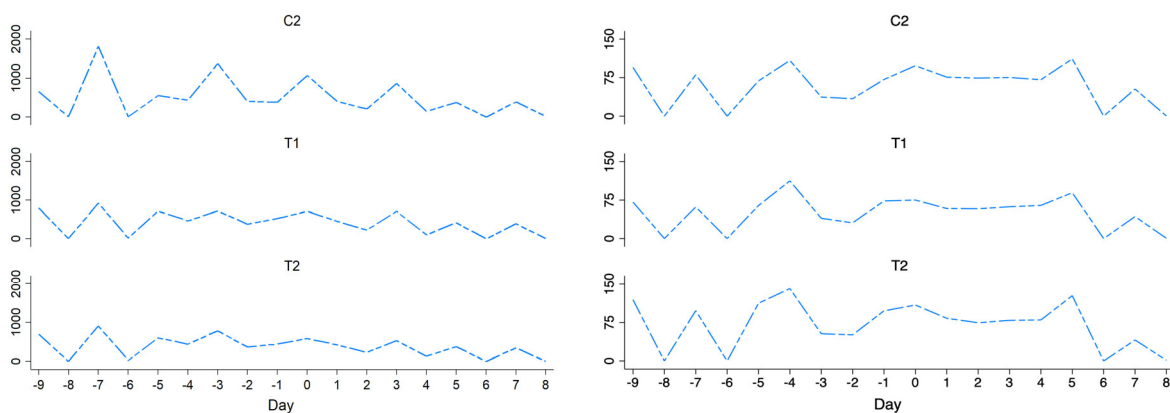
Note: Day 0 is EA day

Internet Appendix 4. Daily experiment activities during the experiment window among treatment groups T1 and T2 and control group C.

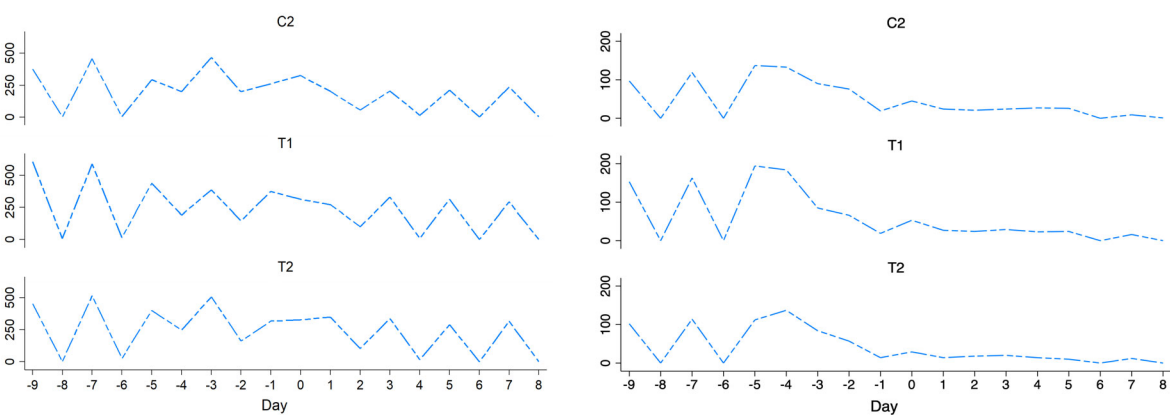
Panel A: Number of total posts (left: China, right: US)



Panel B: Number of reads (in thousands) (left: China, right: US)



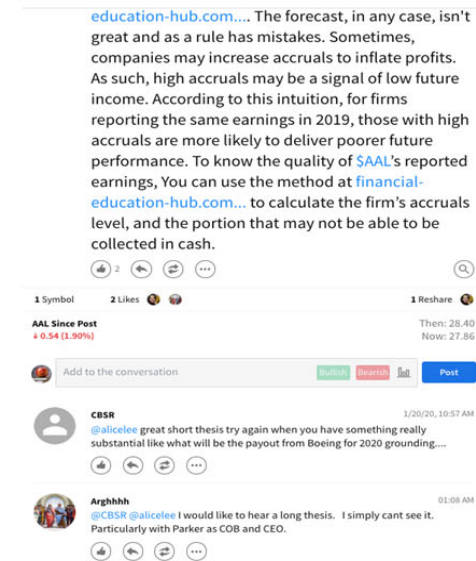
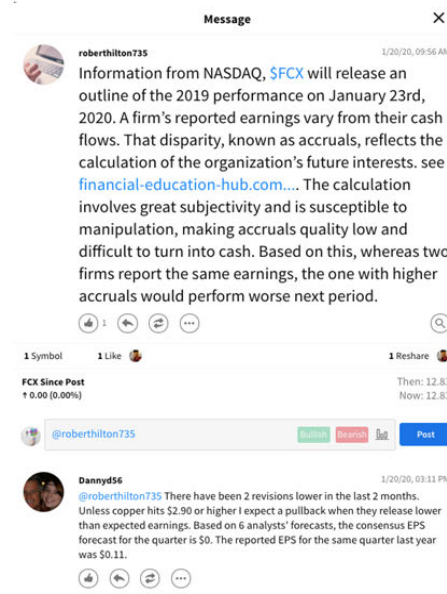
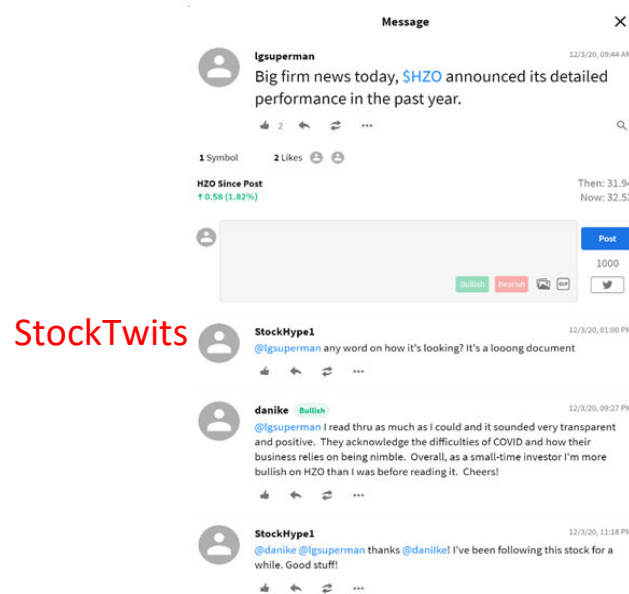
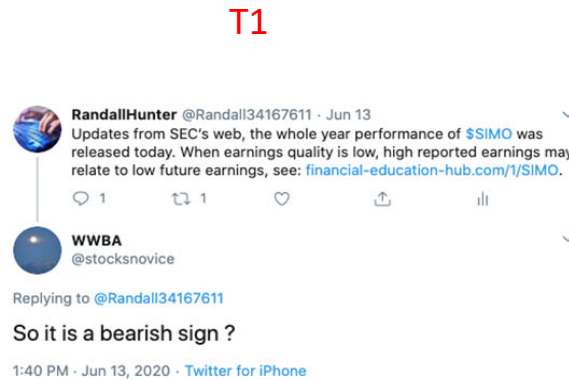
Panel C: Number of comments (left: China, right: US)



Note: Day 0 is the EA day.

Internet Appendix 5. Samples of post responses

Panel A: Twitter and StockTwits



Panel B: Guba EastMoney and XueQiu

Guba

C

张老树83 影响力★★★★★
发布于 2020-03-19 16:14:47 股吧网页版

华电国际将于2020/3/26公布其2019年报

华电国际将于2020/3/26公布其2019年报，各位投资者需要注意了。

评论

全部评论(3)

仍然如此 影响力★★★★★
发布于 2020-03-19 18:36:10
甘肃电投与华电股价差不多时，因着华电业绩预增1倍，结果一念之差，一个拉升，一个被打入地狱，地狱还有十八层，华电A股都看不到升值资金，也许很快3元就是今年最高点了。

股吧热议 影响力★★★★★
发布于 2020-03-19 18:30:28
听里边的高管说这个股今年要去1.3了，后期退市的可能性很大，都快跑吧

仍然如此 影响力★★★★★
发布于 2020-03-19 16:35:53
业绩预告被套了进来，看这架势3元都止不住，H股1元了，拿了5000万支撑股价，若再拿5000万维稳股价就好了。

XueQiu

于老师ip 04-21 12:11 · 来自雪球 阅读 8490

\$京东方A(SZ000725)\$ 京东方A将于2020/4/28公布其2019全年业绩。这里特别提醒各位投资者。

评论...

全部评论 (4)

最后的佛罗纪 04-21 14:58
要跌还是要涨?

炒股涨停板 04-21 14:01
业绩好了哪里

怪蜀黍小巨星 04-21 13:25
感谢

000725粉 04-21 12:48
谢谢提醒

T1

项目的操纵可能违反通用会计准则GAAP。例如，为了提高当期收益，企业可能会选择将费用资本化，而将其排除在库存成本的计算之中。这将导致大量存货成本的估计差错，并导致估计项目的转回与未来现金流不匹配。

估计项目误差也可能不是公司故意制造的。这是基于估计项目不同于后来实际发生的现金流而得出的一种事后特征。例如，在某个时间点，市场需求出人意料地下降，库存不得不以较低的价格出售，甚至报废。在这种情况下，即使没有任何报表操纵的情况下也会出现估计项目和实际发生现金的不匹配。另外，GAAP要求企业以不同于其预期价值的金额进行资产核算，这也将导致在没有任何操纵的情况下出现对估计项目的估计误差。

总之，估计项目代表公司对未来现金流的最佳估计。如果现金流的发生，或如有证据表明预期的现金流不太可能发生，则估计项目和实际现金流不同。第一种情况意味着估计项目准确地预测了未来现金流的发生。第二种情况意味着估计项目估计出现了偏差。估计项目与实际现金流不同。

展开阅读全文

评论

全部评论(3)

八方多美地狂行 发布于 2020-01-23 13:28:12
请问一季度净现金大幅下滑吗?

什么是对的呢 发布于 2020-01-23 13:24:36
辛苦了

代老师zo 05-01 01:32 · 来自雪球 阅读 1716

\$瑞贝卡(SH600439)\$ 瑞贝卡2020/4/30公布了其2019年报，需要注意了。公司报告的利润通常与其现金流不同。这种区别被称为“估计项目”，反映了公司尚未收到现金的利润部分。未收到现金的利润是预估出来的，具有很大的主观性，容易被操纵，使得这部分利润的质量较低，难以转化为现金流入。因此，两...展开

评论...

全部评论 (1)

hmaya 05-01 07:27
应收账款多了近一个亿。。

	4,788,566.00	0.10	14,806,130.00	0.30	-46.76
应收账款	245,334,653.48	4.88	161,582,438.75	3.32	31.84

主要是为了加快应收账款的回收，适当的控制了银行承兑汇票的支付方式。主要是根据市场情况，公司利用应收账款作为客户（或供应商）的担保。

T2

苏老树90 影响力★★★★★
发布于 2020-03-08 11:06:17 股吧网页版

舒神神将于2020/3/17公布其2019年业绩

舒神神将于2020/3/17公布其2019年业绩。这里特别提醒各位投资者。公司报告的利润通常与其现金流不同。这种区别被称为“估计项目”，反映了公司尚未收到现金的利润部分。“估计项目”是公司预估出来的。这个过程会出现有或无的误差。导致这部分利润在未来转变成现金流入的可能性较低。据此，对于那些2019年报告利润相同的公司，那些“估计项目”高的公司更有可能在未来表现不佳。如何判断公司未来业绩，这里给你整理了详细内容，见下图（图里还提供了视频介绍材料的地址）。

评论

股友n08lQm 影响力★★★★★
发布于 2020-03-08 23:40:43
业绩打脸!

股友n08lQm 影响力★★★★★
发布于 2020-03-08 15:56:24
他真正的看点是什

股友n08lQm 发布于 2020-03-08 17:04:23
关键是要做好风险的把控，就看你能力和心态。

股友n08lQm 发布于 2020-03-08 16:46:59
看看财报分析

股友n08lQm 发布于 2020-03-08 16:04:01
只要说2期股价就会飞

股友n08lQm 影响力★★★★★
发布于 2020-03-08 13:50:35
买XQ40的都知道他现在情况，不是买他利润多少，IFRX与本票对冲上53.1美金，为什么?

知行统一 影响力★★★★★
发布于 2020-03-08 15:16:47
老师分析到位，专业!

桃李迎春来发财 影响力★★★★★
发布于 2020-03-08 14:38:20
没看懂，在你说的应该退市了。

daojc 影响力★★★★★
发布于 2020-03-08 13:30:15
这样一看他不得跌价2元以下?

孙老师sk 04-22 01:04 · 来自雪球 阅读 1745

\$太极集团(SH600129)\$ 太极集团将于2020/4/28公布其2019全年业绩。这里特别提醒各位投资者。公司报告利润包括已收到现金的利润和尚未收到现金的利润。估计项目。然而，“估计项目”是公司估算出来的。这种估算可能会有误差甚至错误。比如一些公司甚至利用“估计项目”来操纵报告利润。这导致当期报告利润的“估计项目”部分在未来并不能兑现为实际现金流。为了说明这点，假设公司A和B公告利润相同。A公告利润包含的“估计项目”高于B，可以预测A未来的收益会低于B。如何判断公司未来业绩走向，这里为你准备了相关材料。网页链接。如想估算太极集团所公布利润的质量，可以计算其“估计项目”的具体水平，以及不能兑现成现金流的部分。具体方法参考：网页链接。

评论

全部评论 (1)

掘金大元帅 04-22 08:35
早上卖，下午买没招

Internet Appendix 6. Testing for accrual mispricing

	Chinese market	US market
	(1)	(2)
<i>Accruals</i>	-0.092* (-2.03)	-0.150*** (-2.72)
<i>SUE</i>	0.579** (3.46)	0.013 (1.08)
Industry FEs	Yes	Yes
Observations	13,451	14,243
R-squared	0.085	0.016

Notes: This table reports Fama–MacBeth estimates based on all of the stocks in our sample (stocks in the treatment groups T₁ and T₂, control group C, and comparison group S) from the 2013 to 2018 fiscal years. The dependent variable is *CAR*(11,251), the buy-and-hold return from day 11 to day 251 (day 0 is the EA date) minus the return of the matched portfolio of 5 × 5 size and book-to-market portfolios in the same window. *Accruals* is the change in current assets minus the change in cash holding minus the change in current liabilities plus the change in short-term debt plus depreciation, scaled by total assets. *SUE* measures EA news, which is the announced earnings per share minus the consensus of analyst forecasts in the past year, scaled by the stock price before the EAs. The industry fixed effects are included. Heteroscedasticity and autocorrelation consistent Newey–West (1987) standard error estimates are used. The *t*-statistics are shown in brackets. Significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *, respectively.

Internet Appendix 7. Financial education and accrual pricing: Placebo test

	Chinese market		US market	
	(1)	(2)	(3)	(4)
<i>Treat</i> × <i>Accruals</i>	-0.006		0.009	
	(-0.29)		(0.50)	
<i>Accruals</i>	0.034**		0.037***	
	(2.01)		(3.44)	
<i>Treat</i> × <i>Accruals</i>_{UD}		-0.001		-0.012
		(-0.11)		(-1.35)
<i>Accruals</i> _{UD}		0.035***		0.017**
		(3.45)		(2.12)
<i>SUE</i>	0.039**	0.040**	0.010**	0.011**
	(2.39)	(2.45)	(2.06)	(2.19)
Firm, year and month FEs	Yes	Yes	Yes	Yes
Observations	10,047	10,047	10,689	10,689
R-squared	0.265	0.268	0.485	0.482

Notes: This table reports the effects of our financial education treatment on accrual pricing based on annual EAs for a period before the experiment (from the 2013 to 2018 fiscal years). The sample consists of stocks in the treatment groups T_1 and T_2 and the control group C . The dependent variable is $CAR(0,1)$, the cumulative abnormal return from day 0 to day 1 (day 0 is EA day) based on the market model. *Treat* equals one if a stock is in T_1 or T_2 and zero if it is in C . *Accruals* is the change in current assets minus the change in cash holding minus the change in current liabilities plus the change in short-term debt plus depreciation, scaled by total assets. *Accruals*_{UD} is discretionary accruals in the current year minus discretionary accruals in the previous year. Discretionary accruals are residuals from regressing firms' total accruals (i.e., *Accruals*) on the reciprocal of total assets; the change in sales/total assets; net property, plant, and equipment/total assets; and return on assets in each industry-year. *SUE* measures EA news, which is the announced earnings per share minus the consensus of analyst forecasts in the past year, scaled by the stock price before the EAs. Firm, year, and EA month fixed effects are included in this analysis. The standalone *Treat* is absorbed by the firm fixed effects. The *t*-statistics are shown in brackets. Significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *, respectively.

Internet Appendix 8. Financial education and accrual pricing in the short-term window: Group S as the control

Panel A: Average treatment ($Treat$)				
	Chinese market		US market	
	(1)	(2)	(3)	(4)
$Treat \times Accruals$	-0.048*** (-2.68)		-0.075* (-1.65)	
$Accruals$	0.046*** (2.76)		-0.042 (-1.09)	
$Treat \times Accruals_{UD}$		-0.036* (-1.91)		-0.064*** (-3.27)
$Accruals_{UD}$		0.035** (2.23)		0.013 (0.94)
$Treat$	0.004* (1.67)	0.006*** (2.69)	-0.009*** (-3.07)	-0.004** (-2.26)
SUE	0.336*** (3.65)	0.337*** (3.65)	0.027** (2.19)	0.027** (2.20)
Industry and month FEs	Yes	Yes	Yes	Yes
Observations	1,711	1,711	1,793	1,793
R-squared	0.043	0.042	0.142	0.140
Panel B: Conceptual knowledge treatment (T_I)				
	Chinese market		US market	
	(1)	(2)	(3)	(4)
$T_I \times Accruals$	0.006 (0.24)		-0.025 (-0.51)	
$Accruals$	0.034** (1.99)		-0.019 (-0.51)	
$T_I \times Accruals_{UD}$		-0.016 (-0.71)		-0.075*** (-3.07)
$Accruals_{UD}$		0.026 (1.62)		0.032** (2.44)
T_I	0.009*** (3.28)	0.010*** (3.71)	-0.007** (-2.20)	-0.004** (-2.05)
SUE	0.492*** (4.37)	0.501*** (4.44)	0.020* (1.73)	0.018 (1.49)
Industry and month FEs	Yes	Yes	Yes	Yes
Observations	1,142	1,142	1,194	1,194
R-squared	0.070	0.065	0.175	0.182

Panel C: Conceptual and methodological knowledge treatment (T_2)				
	Chinese market		US market	
	(1)	(2)	(3)	(4)
$T_2 \times Accruals$	-0.061*** (-3.49)		-0.123** (-2.41)	
$Accruals$	0.047*** (2.93)		-0.060 (-1.57)	
$T_2 \times Accruals_{UD}$		-0.049** (-2.27)		-0.093*** (-3.94)
$Accruals_{UD}$		0.037** (2.43)		0.005 (0.40)
T_2	-0.001 (-0.49)	0.002 (0.63)	-0.010*** (-2.99)	-0.002 (-1.12)
SUE	0.185 (1.64)	0.200* (1.77)	0.120*** (2.87)	0.161*** (3.87)
Industry and month FEs	Yes	Yes	Yes	Yes
Observations	1,142	1,142	1,194	1,194
R-squared	0.070	0.040	0.210	0.208

Notes: This table reports the effects of the education treatment on accrual pricing in the short-term window, using group S as the control. The sample in Panel A consists of stocks in the treatment groups T_1 and T_2 and the comparison group S. The sample in Panel B (C) consists of stocks in the treatment groups T_1 (T_2) and the comparison group S. The dependent variable is $CAR(0,1)$, the cumulative abnormal return from day 0 to day 1 (day 0 is EA day) based on the market model. $Treat$ equals one if a stock is in T_1 or T_2 and zero if it is in S. T_1 equals one if a stock is in T_1 and zero if it is in S. T_2 equals one if a stock is in T_2 and zero if it is in S. $Accruals$ is the change in current assets minus the change in cash holding minus the change in current liabilities plus the change in short-term debt plus depreciation, scaled by total assets. $Accruals_{UD}$ is discretionary accruals in the current year minus discretionary accruals in the previous year. Discretionary accruals are residuals from regressing firms' total accruals (i.e., $Accruals$) on the reciprocal of total assets; the change in sales/total assets; net property, plant, and equipment/total assets; and return on assets in each industry-year. SUE measures EA news, which is the announced earnings per share minus the consensus of analyst forecasts in the past year, scaled by the stock price before the EAs. The fixed effects of industry and EA months are included. The t -statistics are shown in brackets. Significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *, respectively.

Internet Appendix 9. Financial education and accrual pricing in a long-term window: Group S as the control

Panel A: Average treatment ($Treat$)				
	Chinese market		US market	
	(1)	(2)	(3)	(4)
$Treat \times Accruals$	0.336* (1.85)		0.094 (0.28)	
$Accruals$	-0.435*** (-2.58)		-0.370 (-1.31)	
$Treat \times Accruals_{UD}$		0.702*** (3.64)		0.226 (1.57)
$Accruals_{UD}$		-0.546*** (-3.45)		-0.055 (-0.55)
$Treat$	0.004 (0.15)	-0.011 (-0.49)	0.006 (0.28)	-0.004 (-0.32)
SUE	0.220 (0.24)	0.307 (0.33)	-0.121 (-1.35)	-0.103 (-1.15)
Industry and month FEs	Yes	Yes	Yes	Yes
Observations	1,711	1,711	1,785	1,785
R-squared	0.098	0.101	0.195	0.195
Panel B: Conceptual knowledge treatment (T_I)				
	Chinese market		US market	
	(1)	(2)	(3)	(4)
$T_I \times Accruals$	-0.338 (-1.35)		-0.401 (-1.01)	
$Accruals$	-0.495*** (-2.86)		-0.183 (-0.62)	
$T_I \times Accruals_{UD}$		0.587*** (2.65)		0.313 (1.63)
$Accruals_{UD}$		-0.630*** (-3.89)		-0.150 (-1.44)
T_I	0.047* (1.70)	0.039 (1.47)	0.001 (0.04)	0.013 (0.81)
SUE	0.865 (0.77)	0.773 (0.68)	-0.119 (-1.31)	-0.096 (-1.04)
Industry and month FEs	Yes	Yes	Yes	Yes
Observations	1,142	1,142	1,187	1,187
R-squared	0.131	0.119	0.174	0.173

Panel C: Conceptual and methodological knowledge treatment (T_2)				
	Chinese market		US market	
	(1)	(2)	(3)	(4)
$T_2 \times Accruals$	0.415** (2.25)		0.550 (1.48)	
$Accruals$	-0.399** (-2.35)		-0.572** (-2.07)	
$T_2 \times Accruals_{UD}$		0.930*** (4.12)		0.385** (2.25)
$Accruals_{UD}$		-0.477*** (-3.02)		-0.011 (-0.11)
T_2	-0.038 (-1.40)	-0.054** (-2.11)	0.014 (0.60)	-0.021 (-1.45)
SUE	-1.237 (-1.05)	-1.276 (-1.09)	-0.472 (-1.56)	-0.537* (-1.78)
Industry and month FEs	Yes	Yes	Yes	Yes
Observations	1,142	1,142	1,191	1,191
R-squared	0.123	0.132	0.269	0.271

Notes: This table reports the effects of education treatment on accrual pricing in the long-term window, using group S as the control. The sample in Panel A consists of stocks in the treatment groups T_1 and T_2 and the comparison group S. The sample in Panel B (C) consists of stocks in the treatment groups T_1 (T_2) and the comparison group S. The dependent variable is $CAR(11,251)$, the buy-and-hold return from day 11 to day 251 (day 0 is the EA date) minus the return of the matched portfolio of 5×5 size and book-to-market portfolios in the same window. $Treat$ equals one if a stock is in T_1 or T_2 and zero if it is in S. T_1 equals one if a stock is in T_1 and zero if it is in S. T_2 equals one if a stock is in T_2 and zero if it is in S. $Accruals$ is the change in current assets minus the change in cash holding minus the change in current liabilities plus the change in short-term debt plus depreciation, scaled by total assets. $Accruals_{UD}$ is discretionary accruals in the current year minus discretionary accruals in the previous year. Discretionary accruals are residuals from regressing firms' total accruals (i.e., $Accruals$) on the reciprocal of total assets; the change in sales/total assets; net property, plant, and equipment/total assets; and return on assets in each industry-year. SUE measures EA news, which is the announced earnings per share minus the consensus of analyst forecasts in the past year, scaled by the stock price before the EAs. The fixed effects of industries and EA months are included. The t -statistics are shown in brackets. Significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *, respectively.

Internet Appendix 10. Financial education effects and reading volume: Group S as the control

Panel A: Average treatment (<i>Treat</i>)						
	Chinese market			US market		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Treat</i> × <i>Accruals</i> _{UD} × <i>Reads</i>	-0.152*** (-6.68)			-0.094*** (-3.10)		
<i>Reads</i>	0.012*** (3.89)			0.011*** (4.72)		
<i>Treat</i> × <i>Accruals</i> _{UD} × <i>CSLS</i>		-0.146*** (-6.85)			-0.059* (-1.92)	
<i>CSLS</i>		0.004 (1.38)			-0.001 (-0.36)	
<i>Treat</i> × <i>Accruals</i> _{UD} × <i>WebView</i>			-0.122*** (-4.57)			-0.082*** (-2.84)
<i>WebView</i>			0.007** (2.25)			-0.001 (-0.55)
<i>Treat</i> × <i>Accruals</i> _{UD}	0.066*** (2.72)	0.044** (2.00)	-0.009 (-0.47)	-0.005 (-0.17)	-0.030 (-1.07)	-0.025 (-1.03)
<i>Accruals</i> _{UD}	0.033** (2.12)	0.034** (2.18)	0.033** (2.11)	0.013 (0.94)	0.014 (0.99)	0.015 (1.06)
<i>Treat</i>	-0.002 (-0.77)	0.003 (1.17)	0.003 (1.33)	-0.011*** (-4.77)	-0.003 (-1.22)	-0.003 (-1.33)
<i>SUE</i>	0.342*** (3.78)	0.335*** (3.68)	0.324*** (3.53)	0.026** (2.08)	0.030** (2.40)	0.022* (1.75)
Industry and month FEs	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,711	1,711	1,711	1,793	1,793	1,793
R-squared	0.078	0.069	0.057	0.153	0.143	0.145
Panel B: Conceptual knowledge treatment (<i>T_I</i>)						
	Chinese market			US market		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>T_I</i> × <i>Accruals</i> _{UD} × <i>Reads</i>	-0.209*** (-6.50)			-0.005 (-0.11)		
<i>Reads</i>	0.008* (1.75)			0.002 (0.52)		
<i>T_I</i> × <i>Accruals</i> _{UD} × <i>CSLS</i>		-0.177*** (-6.02)			-0.171*** (-3.89)	
<i>CSLS</i>		0.001 (0.23)			0.003 (0.75)	
<i>T_I</i> × <i>Accruals</i> _{UD} × <i>WebView</i>			-0.083** (-2.39)			-0.141*** (-3.29)
<i>WebView</i>			0.011** (2.56)			0.000 (0.14)
<i>Treat</i> × <i>Accruals</i> _{UD}	0.129*** (4.13)	0.075*** (2.83)	0.007 (0.30)	-0.071* (-1.79)	0.039 (1.01)	0.003 (0.08)
<i>Accruals</i> _{UD}	0.025 (1.56)	0.026* (1.65)	0.024 (1.50)	0.032** (2.43)	0.031** (2.37)	0.034** (2.57)
<i>T_I</i>	0.004 (0.80)	0.009** (2.30)	0.005 (1.58)	-0.005* (-1.73)	-0.006* (-1.84)	-0.005 (-1.58)
<i>SUE</i>	0.514*** (4.65)	0.485*** (4.36)	0.479*** (4.25)	0.018 (1.50)	0.026** (2.16)	0.008 (0.67)

Industry and month FEs	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,142	1,142	1,142	1,194	1,194	1,194
R-squared	0.106	0.095	0.076	0.182	0.193	0.190
Panel C: Conceptual and methodological knowledge treatment (T_2)						
	Chinese market			US market		
	(1)	(2)	(3)	(4)	(5)	(6)
$T_2 \times Accruals_{UD} \times Reads$	-0.099*** (-3.04)			-0.191*** (-4.77)		
<i>Reads</i>	0.013*** (3.53)			0.015*** (4.66)		
$T_2 \times Accruals_{UD} \times CSLS$		-0.105*** (-3.34)			0.016 (0.41)	
<i>CSLS</i>		0.009** (2.33)			-0.006* (-1.76)	
$Treat \times Accruals_{UD} \times WebView$			-0.073** (-2.00)			-0.096** (-2.44)
<i>WebView</i>			0.010** (2.43)			-0.001 (-0.46)
$T_2 \times Accruals_{UD}$	0.014 (0.45)	0.012 (0.44)	-0.029 (-1.26)	0.025 (0.73)	-0.107*** (-3.24)	-0.053* (-1.83)
$Accruals_{UD}$	0.035** (2.33)	0.035** (2.33)	0.035** (2.32)	0.005 (0.35)	0.009 (0.64)	0.007 (0.52)
T_2	-0.007** (-2.02)	-0.003 (-1.03)	-0.002 (-0.60)	-0.011*** (-4.01)	0.003 (0.75)	-0.001 (-0.47)
<i>SUE</i>	0.184 (1.64)	0.191* (1.70)	0.191* (1.70)	0.150*** (3.61)	0.162*** (3.86)	0.176*** (4.21)
Industry and month FEs	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,142	1,142	1,142	1,194	1,194	1,194
R-squared	0.060	0.055	0.049	0.231	0.210	0.213

Notes: The sample in Panel A consists of stocks in the treatment groups T_1 and T_2 and the comparison group S. The sample in Panel B (C) consists of stocks in the treatment groups T_1 (T_2) and the comparison group S. The dependent variable is $CAR(0,1)$, the cumulative abnormal return from day 0 to day 1 (day 0 is EA day) based on the market model. *Treat* equals one if a stock is in T_1 or T_2 and zero if it is in S. T_1 equals one if a stock is in T_1 and zero if it is in S. T_2 equals one if a stock is in T_2 and zero if it is in S. $Accruals_{UD}$ is discretionary accruals in the current year minus discretionary accruals in the previous year. Discretionary accruals are residuals from regressing firms' total accruals (i.e., *Accruals*) on the reciprocal of total assets; the change in sales/total assets; net property, plant, and equipment/total assets; and return on assets in each industry-year. *Reads* is set to one if the total number of posts read across all of the social media platforms for a stock exceeds the bottom tercile, and zero otherwise. *CSLS* is set to one if the combined number of comments or replies on our posts for a specific stock across all platforms exceeds the bottom tercile or if the posts are "retweeted/shared," "liked," or "saved" on any platforms, and zero otherwise. *WebView* is set to one if the number of clicks on both type 1 and type 2 link pages for a stock surpasses the bottom tercile among the treatment stocks, and zero otherwise. *SUE* is earnings per share minus the consensus of analyst forecasts in the past year, scaled by the stock price before the EAs. The interaction terms between the engagement measures and *Treat* and $Accruals_{UD}$ are omitted to avoid multicollinearity because the engagement variable indicates a subgroup of the treatment stocks and takes the value of zero in all of the control stocks (stocks in group S do not receive any intervention). The *t*-statistics are shown in brackets. Significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *, respectively.

Internet Appendix 11. Financial education effects and retail investors: Group S as the control

Panel A: Average treatment ($Treat$)

	Chinese market	US market
	(1)	(2)
$Treat \times Accruals_{UD} \times RI$	-0.021 (-0.73)	-0.185*** (-3.08)
$Treat \times RI$	0.010 (1.52)	0.012** (2.46)
$Accruals_{UD} \times RI$	-0.009 (-0.31)	0.204*** (4.30)
RI	-0.017*** (-3.39)	-0.006 (-1.39)
$Treat \times Accruals_{UD}$	-0.015 (-0.72)	0.106* (1.89)
$Accruals_{UD}$	0.035 (1.08)	-0.175*** (-3.81)
$Treat$	0.004 (1.14)	-0.014*** (-3.10)
SUE	0.343** (2.27)	0.026** (2.16)
Industry and month FEs	Yes	Yes
Observations	1,711	1,793
R-squared	0.058	0.153

Panel B: Conceptual knowledge treatment (T_I)

	Chinese market	US market
	(1)	(2)
$T_I \times Accruals_{UD} \times RI$	-0.105*** (-3.28)	-0.156** (-2.30)
$T_I \times RI$	0.001 (0.06)	0.004 (0.64)
$Accruals_{UD} \times RI$	0.001 (0.04)	0.174*** (3.88)
RI	-0.019*** (-3.51)	-0.005 (-1.34)
$T_I \times Accruals_{UD}$	0.062* (2.07)	0.067 (1.08)
$Accruals_{UD}$	0.022 (0.56)	-0.129*** (-2.96)
T_I	0.015 (1.71)	-0.007 (-1.39)
SUE	0.552*** (4.62)	0.017 (1.47)
Industry and month FEs	Yes	Yes
Observations	1,142	1,194
R-squared	0.106	0.194

Panel C: Conceptual and methodological knowledge treatment (T_2)

	Chinese market	US market
	(1)	(2)
$T_2 \times Accruals_{UD} \times RI$	0.035 (0.67)	-0.206*** (-3.07)
$T_2 \times RI$	0.015*** (4.16)	0.014** (2.55)
$Accruals_{UD} \times RI$	-0.011 (-0.45)	0.203*** (4.35)
RI	-0.017*** (-3.99)	-0.003 (-0.68)
$T_2 \times Accruals_{UD}$	-0.066 (-1.62)	0.096 (1.56)
$Accruals_{UD}$	0.038 (1.34)	-0.181*** (-4.00)
T_2	-0.003* (-1.92)	-0.013*** (-2.67)
SUE	0.207* (1.81)	0.156*** (3.79)
Industry and month FEs	Yes	Yes
Observations	1,142	1,194
R-squared	0.059	0.228

Notes: The sample in Panel A consists of stocks in the treatment groups T_1 and T_2 and comparison group S. The sample in Panel B (C) consists of stocks in the treatment groups T_1 (T_2) and the comparison group S. The dependent variable is $CAR(0,1)$, the cumulative abnormal return from day 0 to day 1 (day 0 is the EA day) based on the market model. $Treat$ equals one if a stock is in T_1 or T_2 and zero if it is in S. T_1 equals one if a stock is in T_1 and zero if it is in S. T_2 equals one if a stock is in T_2 and zero if it is in S. $Accruals_{UD}$ is discretionary accruals in the current year minus discretionary accruals in the previous year. Discretionary accruals are residuals from regressing firms' total accruals (i.e., $Accruals$) on the reciprocal of total assets; the change in sales/total assets; net property, plant, and equipment/total assets; and return on assets in each industry-year. RI is assigned a value of one if the intensity of retail investors exceeds the bottom tercile of our sample, and zero otherwise. We measure the intensity of retail investors using 100% minus the number of shares held by institutional investors over the total number of tradable shares. SUE measures the EA surprise, which is the announced earnings per share minus the consensus of analyst forecasts in the past year, scaled by the stock price before the EAs. The fixed effects of industries and EA months are included. The t -statistics are shown in brackets. Significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *, respectively.

Internet Appendix 12. Compare control and comparison groups

	Chinese market		US market	
	(1)	(2)	(3)	(4)
<i>Treat</i> × <i>Accruals</i>	0.013		-0.076	
	(1.46)		(-1.43)	
<i>Accruals</i>	0.004		0.001	
	(0.54)		(0.03)	
<i>Treat</i> × <i>Accruals</i>_{UD}		0.025		-0.007
		(1.34)		(-0.26)
<i>Accruals</i> _{UD}		0.030*		0.011
		(1.86)		(0.95)
<i>Treat</i>	0.015***	0.013***	-0.003	0.001
	(6.15)	(5.44)	(-0.94)	(0.26)
<i>SUE</i>	0.629***	0.582***	0.170**	0.172**
	(5.60)	(5.23)	(2.41)	(2.43)
Industry and month FEs	Yes	Yes	Yes	Yes
Observations	1,146	1,146	1,189	1,189
R-squared	0.118	0.138	0.165	0.163

Notes: The sample consists of stocks in the control group C and the comparison group S in our experiment period. The dependent variable is $CAR(0,1)$, the cumulative abnormal return from day 0 to day 1 (day 0 is EA day) based on the market model. *Treat* equals one if a stock is in group C and zero if it is in group S. *Accruals* is the change in current assets minus the change in cash holding minus the change in current liabilities plus the change in short-term debt plus depreciation, scaled by total assets. *Accruals*_{UD} is discretionary accruals in the current year minus discretionary accruals in the previous year. Discretionary accruals are residuals from regressing firms' total accruals (i.e., *Accruals*) on the reciprocal of total assets; the change in sales/total assets; net property, plant, and equipment/total assets; and return on assets in each industry-year. *SUE* measures EA news, which is the announced earnings per share minus the consensus of analyst forecasts in the past year, scaled by the stock price before the EAs. The fixed effects of industries and EA months are included. The *t*-statistics are shown in brackets. Significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *, respectively.

Internet Appendix 13. Compare the pricing of positive and negative accruals

Panel A: Sample of positive accruals (accruals in the top decile)		
	Chinese market	US market
	(1)	(2)
<i>Treat</i> × <i>Accruals</i>	-0.340***	-0.431***
	(-4.29)	(-2.67)
<i>Accruals</i>	0.206***	0.368**
	(4.29)	(2.44)
<i>Treat</i>	0.062***	-0.019**
	(3.52)	(-2.07)
<i>SUE</i>	-0.592	0.928
	(-1.37)	(1.47)
Industry and month FEs	Yes	Yes
Observations	171	180
R-squared	0.428	0.741

Panel B: Sample of negative accruals (accruals in the bottom decile)		
	Chinese market	US market
	(1)	(2)
<i>Treat</i> × <i>Accruals</i>	-0.020	-0.098
	(-1.45)	(-0.04)
<i>Accruals</i>	0.008	-0.554
	(0.84)	(-0.35)
<i>Treat</i>	-0.023**	-0.011
	(-2.34)	(-0.02)
<i>SUE</i>	0.216	0.300
	(0.52)	(0.70)
Industry and month FEs	Yes	Yes
Observations	170	182
R-squared	0.301	0.476

Notes: The sample consists of stocks with high and low accruals in the treatment groups T_1 and T_2 and the control group C in our experiment period. The dependent variable is $CAR(0,1)$, the cumulative abnormal return from day 0 to day 1 (day 0 is EA day) based on the market model. *Treat* equals one if a stock is in T_1 or T_2 and zero if it is in C . *Accruals* is the change in current assets minus the change in cash holding minus the change in current liabilities plus the change in short-term debt plus depreciation, scaled by total assets. *SUE* measures EA news, which is the announced earnings per share minus the consensus of analyst forecasts in the past year, scaled by the stock price before the EAs. The fixed effects of industries and EA months are included. Panel A reports the estimates based on a sample of positive accruals (*Accruals* on the top decile). Panel B reports the estimates based on a sample of negative accruals (*Accruals* on the bottom decile). The *t*-statistics are shown in brackets. Significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *, respectively.

Internet Appendix 14. Financial education effects in the post-experiment period

Panel A: Baseline		
	Chinese market	US market
	(1)	(3)
<i>Treat</i> × <i>Accruals_{UD}</i>	-0.096***	-0.027*
	(-3.22)	(-1.73)
<i>Accruals_{UD}</i>	0.087***	0.036***
	(4.20)	(2.60)
<i>Treat</i>	0.006***	0.007***
	(2.83)	(2.84)
<i>SUE</i>	0.009	0.155***
	(0.37)	(2.61)
Industry and Year FEs	Yes	Yes
Observations	1,679	1,693
R-squared	0.063	0.159

Panel B: Conceptual vs. methodological education				
	Chinese market		US market	
	Treat = <i>T₁</i>	Treat = <i>T₂</i>	Treat = <i>T₁</i>	Treat = <i>T₂</i>
	(1)	(2)	(3)	(4)
<i>Treat</i> × <i>Accruals_{UD}</i>	-0.130***	-0.066*	-0.027*	-0.040**
	(-3.54)	(-1.76)	(-1.78)	(-2.03)
<i>Accruals_{UD}</i>	0.099***	0.085***	0.037***	0.039***
	(4.52)	(3.99)	(2.91)	(2.71)
<i>Treat</i>	0.011***	0.002	0.002	0.012***
	(3.93)	(0.67)	(0.82)	(4.10)
<i>SUE</i>	0.025	-0.029	0.176***	0.133*
	(0.83)	(-1.04)	(2.70)	(1.91)
Industry and Year FEs	Yes	Yes	Yes	Yes
Observations	1,116	1,122	1,130	1,128
R-squared	0.128	0.057	0.136	0.244

Notes: This table reports the effects of the education treatment on accrual pricing one year after our experiment (annual earnings announced in 2021). The sample consists of stocks in the treatment groups *T₁* and *T₂* and the control group *C*. The dependent variable is *CAR*(0,1), the cumulative abnormal return from day 0 to day 1 (day 0 is EA day) based on the market model. In Panel A, *Treat* equals one if a stock is in *T₁* or *T₂* and zero if it is in *C*. In Panel B, *Treat* indicates *T₁* or *T₂*. *T₁* equals one if a stock is in *T₁* and zero if it is in *C*. *T₂* equals one if a stock is in *T₂* and zero if it is in *C*. *Accruals* is the change in current assets minus the change in cash holding minus the change in current liabilities plus the change in short-term debt plus depreciation, scaled by total assets. *Accruals_{UD}* is discretionary accruals in the current year minus discretionary accruals in the previous year. Discretionary accruals are residuals from regressing firms' total accruals (i.e., *Accruals*) on the reciprocal of total assets; the change in sales/total assets; net property, plant, and equipment/total assets; and return on assets in each industry-year. *SUE* measures EA news, which is the announced earnings per share minus the consensus of analyst forecasts in the past year, scaled by the stock price before the EAs. The fixed effects of industries and EA months are included. The *t*-statistics are shown in brackets. Significance at the 1%, 5%, and 10% levels is indicated by ***, **, and *, respectively.